

Master Thesis

Grammatical Gender Prediction of English Loanwords in German Using Transformer-Based Language Models

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Abstract

This thesis investigates whether bidirectional encoder-transformer models can predict the grammatical gender of English loanwords embedded in German sentences. Grammatical gender in German is expressed through agreement phenomena, most directly through a noun's determiner in the nominative or accusative case. The determiner assigned to a loanword is therefore used as a proxy for its grammatical gender. To evaluate model performance on grammatical gender prediction, a dataset of 989 manually constructed German sentences containing English loanwords in the nominative or accusative case was created. The dataset is organised into four main categories based on established patterns of gender assignment to loanwords in German: morphological mapping, biological gender, translation, and phonological patterns. An additional category of plural forms are included as a control condition.

Four BERT-style models are evaluated on a fill-mask task in both zero-shot and finetuned settings: mBERT, germanBERT, gBERT, and dbmdzBERT. The results split predictions into three types: correct, underspecified and incorrect. For the analysis, alternative predictions that formed grammatical sentences were taken into account (e.g., possessive pronouns).

The results show that finetuning improves performance across all models and most categories, with the largest gains observed for suffix-based morphological categories. Monolingual German models outperform the multilingual mBERT in both settings, suggesting that language-specific pretraining provides better alignment with German grammatical gender. Morphological suffix-based cues are reliably captured, while translation-based categories remain the most challenging. Tokenisation patterns appear to influence performance, as gender-marking suffixes, when represented as individual subtokens, are more consistently linked to correct gender predictions. Loanword frequency and length show limited to no effects on model performance.

Declaration of Authorship

I, Hannah Altgassen, declare that this thesis, titled *Grammatical Gender Prediction of English Loanwords in German Using Transformer-Based Language Models* and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a Master degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Date: 26.06.2026

Signed: 

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Chapter 1

Introduction

This thesis focuses on the grammatical gender assignment of bidirectional encoder transformer models to English loanwords embedded in German sentences. In German, grammatical gender is not marked directly on nouns alone but is typically expressed through agreement phenomena, for example through definite and indefinite determiners as well as adjective and pronoun inflection. As a result, determiners can provide a reliable surface-level indicator of a noun's grammatical gender. In this study, the determiner assigned to a loanword is therefore used as a proxy for its grammatical gender, a method that will be further explained in Section 2. The choice of the correct determiner with respect to number and gender is essential for producing grammatically well-formed sentences in German.

In recent years, the use of English vocabulary in German has increased significantly, and native speakers commonly integrate English vocabulary into German syntactic structures (Cantarini, 2023). In such cases, determiner choice, and thus gender assignment, typically occur automatically and without conscious effort, although likely still guided by systematic linguistic patterns (as discussed in Chapter 2.3). These patterns allow speakers to assign grammatical gender even to newly borrowed words.

Given that native speakers appear to assign grammatical gender to novel foreign words with ease, this thesis investigates whether bidirectional transformer models are capable of capturing and reproducing these patterns of gender assignment.

1.1 Motivation

Grammatical gender is a fundamental aspect of many languages, including German. Unlike lexical meaning, the gender of nouns is often arbitrary and must be learned as part of a word's grammatical properties. This becomes especially interesting in the case of loanwords, where speakers must assign gender to lexical items not part of a languages original vocabulary. English loanwords in German provide a particularly interesting test case, as English does not have grammatical gender, meaning that any gender assignment must arise from adaptation strategies within German rather than from properties inherited from the source language.

Understanding how grammatical gender is assigned to loanwords is not only relevant for theoretical linguistics but also for computational linguistics. Many NLP systems, particularly transformer-based encoder models, are increasingly used in multilingual settings where linguistic phenomena such as loanwords or code-switching naturally occur. However, the question of whether these models learn grammatical patterns or

whether their performance is driven by surface-level correlations in training data remains. Therefore, this thesis will focus on the topic of gender assignment to loanwords.

1.2 Research Question and Hypothesis

This thesis aims to investigate the ability of bidirectional encoder models to predict the grammatical gender of English loanwords embedded in German sentences. The determiner of a noun will be used as a proxy for gender. The models will be predicting determiners both in a zero-shot setting and after fine-tuning on a manually created dataset. Three monolingual German models and one multilingual model are used for comparison across the evaluated settings.

The thesis aims to answer the following research question:

1. Can pretrained bidirectional encoder models predict the grammatical gender of English loanwords embedded into German sentences, and can finetuning improve their performance?

Furthermore, the following subquestions are investigated:

SQ.1 Do multilingual models outperform monolingual models?

SQ.2 Which underlying patterns are captured by language models?

To answer these questions, the following four models will be evaluated on a fill-mask task in which they predict the definite determiner for English loanwords embedded in German sentences:

- google-bert/bert-base-multilingual-cased (mBERT) (Google, 2024)
- google-bert/bert-base-german-cased (germanBERT) (Google, 2019)
- deepset/gbert-base (gBERT) (deepset, 2020)
- dbmdz/bert-base-german-cased (dbmdzBERT) (dbmdz, 2026)

This setup allows for a comparison of how well each model captures grammatical gender assignment in both a zero-shot and a finetuned setting. By analysing the predictions, the study assesses both overall performance and differences between multilingual and monolingual models.

It is hypothesised that fine-tuning will improve performance in gender assignment when using determiners as a proxy for grammatical gender. This is in line with the findings of Devlin et al. (2018), who claim that BERT-style models achieve very good performance on downstream tasks through task-specific fine-tuning, without requiring task-specific architectures. Furthermore, it is expected that the multilingual model will outperform monolingual models due to its pretraining on both German and English, which allows for cross-linguistic transfer and shared representations across languages (Pires et al., 2019). It is hypothesised that the availability of shared representations may be particularly helpful for translation-based instances, because it captures word meaning in both languages, which could lead to better alignment between translations.

1.3 Chapter Outline

This thesis is structured into eight chapters. Section 2 provides the theoretical background, including an introduction to grammatical gender assignment in German, gender assignment to loanwords, transformer model architectures, potential model biases, and relevant previous research. Section 3 introduces the manually constructed dataset created for this thesis. Section 4 outlines the methodological approach, while Section 5 presents the results. Finally, the analysis (Section 6) and discussion (Section 7) evaluate the models' abilities to assign grammatical gender to English loanwords in German sentences.

Chapter 2

Literature Review

This Chapter introduces the linguistic and computational background on which the experimental setup is grounded. First, gender assignment in German is briefly discussed, followed by an introduction to anglicisms and the patterns used to assign gender to English loanwords in German. Furthermore, the computational modelling of gender in word embeddings is examined. This is followed by an introduction to transformer-based language models and the four models used in the current experimental setup. Lastly, prior studies on grammatical gender prediction are discussed.

2.1 Grammatical Gender in German

There are three grammatical genders in German: masculine, feminine and neuter. The gender of a noun can be reflected in alternative words in the sentence, such as the determiner, adjectives, and (possessive or demonstrative) pronouns. The masculine gender is the default or "unmarked" gender in the German language (Virtanen, 2023). For standalone nouns, German assigns the following definite determiners in the nominative case: *der* (masculine), *die* (feminine), and *das* (neuter).

Additionally, German has case marking. Determiners, therefore, inflect according to grammatical case, number, and gender. The full set of definite determiners across cases and genders is displayed in Table 2.1. In some cases, different genders share identical surface forms. Therefore, this thesis will primarily focus on the nominative forms *der, die,* and *das*, and the accusative forms, as they directly reflect the gender of a noun and allow for clearer comparisons across model predictions and the reference standard.

Gender	Nominative	Accusative	Dative	Genitive
Masculine	der	den	dem	des
Feminine	die	die	der	der
Neuter	das	das	dem	des

Table 2.1: German Definite Articles by Gender and Case

An example of a standard German sentence can be seen in Example 1. Here, the primary indicator of grammatical gender is the direct neuter determiner in the nominative case (*das*). However, this determiner can also be replaced by an alternative determiner (e.g., possessive or demonstrative forms mentioned above) while still forming a grammatically correct sentence (see Example 2) and indicating grammatical gender.

Possible alternatives that indicate grammatical gender will be taken into account when analysing the results (see Section 5.1).

(1) Das Essen ist lecker
the.DEF.N.NOM.SG food.N.SG be.3SG delicious
'The food is delicious.'

(2) Mein Essen ist lecker
my.POSS.1SG.N.NOM.SG food.N.SG be.3SG delicious
'My food is delicious.'

Previous research has identified several patterns by which grammatical gender is assigned in German. These include semantic groupings, morphological and phonological patterns, and biological gender principles, and will be discussed in the following sections.

2.1.1 Morphological Patterns

The first mechanism through which grammatical gender is assigned in German is morphology. The primary pattern is suffixes that strongly correlate with a noun's gender. The full range of suffixes corresponding to each of the three genders is displayed in Table 2.2 (Oleschko, 2010).

Gender	**Suffixes**
Masculine	-er, -ler, -ner, -ling, -and, -ant/-ent, -är, -eur (-ör), -ier, -ismus, -ist, -(at)or
Feminine	-e, -ei, -heit/-(ig)keit, -in, -schaft, -ung, -ade, -age, -anz/-enz, -erie, -ie, -ik, -(at)ion, -ität, -(at)ur
Neutral	-chen, -lein, -sel, -tum

Table 2.2: German Noun Suffixes by Grammatical Gender

The suffixes displayed in Table 2.2 almost always correspond to the gender they are listed under. They are highly predictive of grammatical gender in German derivational morphology; however, they do not constitute absolute rules. Rather, they reflect strong statistical regularities with a limited number of exceptions (Oleschko, 2010).

2.1.2 Phonological Patterns

Phonological cues are also relevant for the assignment of grammatical gender to German nouns, though they usually rank secondary to semantic and morphological rules (see Section 2.1.5). There are differences between monosyllabic and polysyllabic nouns, as most phonological effects are especially strong for monosyllabic nouns.

For monosyllabic nouns, it is proposed that phonological structure influences gender assignment in a probabilistic rather than deterministic way (Köpcke and Zubin, 1984). Köpcke and Zubin identify several tendencies linking phonological properties of word form to grammatical gender. These include patterns related to word-initial and word-final consonant structures. For example, nouns beginning with consonant clusters such as *kn-* tend to be more frequently masculine. Similarly, nouns ending in consonant clusters with a non-sibilant fricative followed by a plosive (e.g. *-ft, -cht*) show a tendency for the feminine gender (Köpcke and Zubin, 1984).

Furthermore, German words beginning with plosives are more likely to be masculine, and nouns ending in consonant clusters also tend to be masculine (Köpcke and Zubin, 1984). The phonological patterns are seen as statistical tendencies, rather than absolute rules. Importantly, phonological effects are generally considered to operate only when morphological and semantic cues do not provide a clear gender assignment (Köpcke and Zubin, 1984).

2.1.3 Semantic Patterns

Gender Assignment in German can also be led by semantic principles. These can be split into two types of groupings: biological gender patterns and the *Gattungsprinzip* (English: Genus Principles). These are introduced in this section.

Biological Gender Patterns

An important mechanism of gender assignment in German is the biological gender principle (German: *Genus-Sexus-Prinzip*). Though German distinguishes between biological gender (*Sexus*) and grammatical gender (*Genus*) nouns referring to individuals with a biological gender (such as humans and animals), typically receive grammatical gender corresponding to their *Sexus* (Helbig and Buscha, 1996). Nouns referring to male individuals are assigned the masculine gender, and nouns referring to female individuals receive the feminine gender. For example, *der Mann* (English: "the man") or *der Sohn* (English: "the son") refer to male individuals and receive the masculine gender expressed, for example, through the masculine nominative definite determiner *der* (Helbig and Buscha, 1996). *Die Frau* (English: "the woman") and *die Tochter* (English: "the daughter"), on the other hand, refer to female individuals and receive the feminine gender expressed, for example, through the feminine nominative definite determiner *die* (Helbig and Buscha, 1996).

An exception are young human or animal individuals. They typically receive the neuter gender, no matter their biological gender. For example, *das Baby* (English: "the baby") or *das Lamm* (English: "the lamb"), receive the neuter gender expressed, for example, through the nominative neuter determiner *das* (Dudenredaktion, 2009). This also applies to diminutive forms such as *das Mädchen* (English: "the Girl"), where the neuter grammatical gender overrides the biological gender of the referent.

Semantic Groupings (Genus Principle)

Semantic Patterns are another category by which the gender of a word can be distinguished in German. According to Helbig and Buscha (1991), nouns belonging to categories such as seasons, months, days of the week, cardinal directions, winds, precipitation, car brands, as well as minerals and rocks are typically masculine. In contrast, nouns referring to flowers, trees, and numbers tend to receive feminine determiners. Neuter determiners are commonly assigned to nouns in categories such as hotels, cafés, chemical and physical elements, letters, colours, diminutives, and languages (Oleschko, 2010). Although these semantic category patterns are largely reliable, they are not absolute rules. Rather, they serve as descriptive generalisations that are primarily used to illustrate tendencies and facilitate gender assignment for L2 learners of German.

2.1.4 Plurals

Plural formation in German mainly uses the following four suffixes: *-e, -en, -er* and *-s*. Nouns that are feminine in the singular form take the *-en* suffix in about 96.6% of cases (Wegener, 1999). Weak masculines take the *-en* suffix in all cases, while strong masculines and neuters use the *-e* suffix to form their regular plural in 70.2% of cases (Wegener, 1999). Though the gender of German nouns does not change when they appear in plural forms all plurals take the determiner *die*, which is also used for singular female nouns in both the nominative and the accusative case (Wegener, 1999).

2.1.5 Hierarchy for Gender Assignment in German

Gender assignment in German follows a hierarchy of strategies. Biological gender overrides other gender assignment rules. Furthermore, morphological cues (such as suffixes) take priority, followed by semantic and phonological cues when morphological information is absent or ambiguous (Nyenhuis and Pittner, 2012; Oleschko, 2010). A well-known exception to this is diminutive forms such as *das Mädchen*, where the neuter grammatical gender is used for a feminine referent, and pronoun usage may vary between *sie* (English: "she") and *es* (English: "it").

2.1.6 Additional Notes on Gender Assignment in German

German nouns do not involve a translation-based category for gender assignment, as they are inherently part of the language and therefore do not require mapping from a source language. They are assigned grammatical gender solely through the patterns introduced above, such as morphological, phonological, semantic, and biological gender principles. This distinction is important because translation-based gender assignment becomes relevant in the context of loanwords, where gender may be influenced by the corresponding form or meaning in another language. Anglicisms and Loanwords will be discussed in the following section.

2.2 Anglicisms in German

English words that are embedded into other languages are formally known as anglicisms. Both loanwords and foreign words are considered anglicisms. The differences between the two will be explained in Section 2.2.1. This thesis, however, primarily focuses on loanwords rather than foreign words, as these forms are more integrated into the language and therefore follow its general grammatical and morphological patterns more consistently.

2.2.1 Loanwords vs Foreign Words

The use of English words in other languages has increased in recent years. However, it is important to distinguish between loanwords (*Lehnwörter*) and foreign words (*Fremdwörter*). All words adopted from another language are initially classified as foreign words. These often show non-native sound–spelling relationships. For example, in the word *make-up*, German speakers may pronounce the /u/ as an /a/, or the word may contain phonological or orthographic patterns that are unusual in German (e.g. "Job" /dʒ/). In some cases, foreign words may also keep part of their original

grammatical structure. For example, the English plural *-s* suffix is often taken over for anglicisms used in German (Duden, 2026). For example, the plural of *Jobs* will remain *die Jobs* (neuter) in German. The German equivalent *die Berufe*, adds an *-e* as the plural marker, as *-s* is not a plural marker in German.

Loanwords are words that were previously foreign words but have been integrated into the German language. This integration includes adaptations such as changes in pronunciation patterns (e.g. *Balkon* → [bal'ko:n]) as well as morphological adaptations. An example of such a morphological adaptation is *die Manager* (English: "the managers"), where plural marking follows German morphological rules, through the addition of the suffix *-er* for plural formation. Furthermore, it entails that foreign spelling patterns are to be replaced by common German spelling (e.g. *Photograph* to *Fotograf*).

The line between loanwords and foreign words is very fine and may be interpreted differently by L1 speakers, especially when the foreign roots of a word are not known (Duden, 2026).

2.3 Gender Assignment to English Loanwords

Anglicisms that are embedded into German predominantly obtain the masculine grammatical gender (Nyenhuis and Pittner, 2012). This is likely due to masculine functioning as the default gender for German nouns. Further research shows that a minority of anglicisms are assigned feminine grammatical gender (approximately 14%), while around 28% receive neuter grammatical gender. The default gender assignment pattern for English loanwords in German is masculine > feminine > neuter (Nyenhuis and Pittner, 2012). However, gender assignment is also shaped by additional patterns such as lexical and morphological factors, which will be discussed in this section.

2.3.1 Competing Theories on Gender Assignment

Several linguists have proposed theories on gender assignment to loanwords. Schulte-Beckhausen (2002) proposes that borrowed nouns tend to take neuter gender as a default. She argues that, when gender cannot be determined through semantic, morphological, or phonological cues, speakers may implicitly associate the loanword with semantically underspecified German nouns such as *das Ding* (English:"the thing") or *das Objekt* (English:"the object"), which are neuter. However, the idea of a neuter default does not match the actual gender distribution as introduced in 2.3.

Other linguists argue that this does not match the distribution of gender across English loanwords, where on average 58% of English loanwords receive the masculine gender, 14% the feminine gender and 28% the neuter gender (Nyenhuis & Pittner, 2012).

2.3.2 Morphological Patterns

The gender assignment of anglicisms follows patterns similar to those observed for German vocabulary. Since English and German belong to the same language family, several noun-forming suffix patterns can be directly translated. Given that gender assignment in German is partly correlated with morphological structure, the corresponding English suffixes trigger similar gender assignment patterns for loanwords. For instance, nouns ending in *-er* are typically assigned masculine grammatical gender in German, although

exceptions exist (e.g. *das Fenster*, English: "the window"). Correspondingly, English loanwords with an *-er* ending tend to be assigned masculine gender (e.g. *der Bestseller*). Another example involves the suffixes *-heit* and *-keit*, which correspond to the English suffix *-ness* (e.g. *die Helligkeit*, English: "the brightness"). Since these suffixes are associated with feminine grammatical gender in German, English *-ness* formations likewise tend to receive feminine gender. Further examples of suffix correspondences are provided in Table 2.3.

English Suffix	**English Example**	**German Suffix**	**German Example**	**Gender for Loanword**
-er	Teacher	-er	Lehrer	der
-ness	Brightness	-heit/-keit	Helligkeit	die
-ing	Reading	-en	Lesen	das
-ion	Action	-ion	Aktion	die
-ity	Publicity	-ität	Popularität	die

Table 2.3: Comparison of English and German Suffixes by Grammatical Gender

English loanwords ending in *-ing* usually get neuter gender in German (see Table 2.3). When words ending in *-ing* are incorporated into German, they are interpreted as nouns denoting activities or processes, i.e. nominalised infinitives. In German, nouns derived from verbs in this way are consistently neuter. For example, *das Laufen* (English:"the walking"), *das Arbeiten* "the working"), and *das Kochen* ("the cooking"). Consequently, German speakers tend to assign neuter gender to English *-ing* forms by analogy with native German nominalisations of verbs. Hence, the neuter gender is primarily assigned to nouns ending in *-ing* due to the nominalisation.

Plurals

English loanwords in German initially retain their original English plural forms. Over time, however, some nouns develop an additional plural variant that conforms to German morphological patterns (Griebel, 2006). Plural formations of loanwords with German plural suffix are possible, but rare (Griebel, 2006). All plural nouns in German, including English loanwords, take the determiner *die*, as gender distinctions are primarily made in the singular number (Krifka, 2000).

2.3.3 Phonological Patterns

Phonological patterns also play a role in the gender assignment for English loanwords, although they generally function as secondary mechanisms compared to morphological cues (Griebel, 2006). In particular, Griebel (2006) identifies three core phonological processes involved in gender assignment: rhyme analogy, phonological similarity, and monosyllabic structural tendencies.

Rhyme analogies are cases where the gender is assigned by analogy to phonologically similar native words, particularly rhyming forms. For example, *der Snack* is patterned after *der Dreck* (English: "the dirt"), *der Fleck* (English: "the spot"), *der Schreck* (English: "the scare") (Griebel, 2006). In addition, gender may be assigned through broader phonological similarity, where loanwords adopt the gender of German words with similar sound structures, even in the absence of semantic overlap (Griebel, 2006).

An example of this would be the English word *Gift*, which is phonologically identical to the German word *das Gift* (English: Poison).

Finally, the overall phonological shape of a word may influence gender assignment: monosyllabic nouns and those ending in consonants, particularly plosives, are frequently assigned masculine gender (Griebel, 2006). Additionally, a large number of masculine loanwords end in voiceless plosives (e.g. /p/, /t/, /k/) and are assigned the masculine gender (Griebel, 2006). This thesis will only take the phonological shape into account when trying to model gender assignment through phonological patterns.

2.3.4 Semantic Patterns

There are three semantic patterns according to which grammatical gender can be assigned to English loanwords. These include the biological gender category (as introduced in Section 2.1.3), translation patterns and the genus principle. The translation and genus principle are observation by Griebel (2006) and Vanhaelemeesch (2020) respectively. They are linguistic patterns that are overridden by the genus principle.

Biological Gender

The biological gender pattern is seen as a subcategory of semantic patterns in previous research. Here, grammatical gender is assigned to loanwords involves reference to biological or social gender (Yang, 2010). Referents associated with the sex "female" are assigned feminine grammatical gender, while those associated with the sex "male" are assigned masculine grammatical gender. Examples include *die Queen* (feminine), English: "the Queen" and *der King* (masculine), English: "the King".

Translation

Another strategy through which grammatical gender is assigned to English loanwords involves the gender of their equivalent German translation, a pattern that is particularly common for monosyllabic forms (Jach, 2011). Examples include *der Trip* (German: "der Ausflug", masculine) and *der Job* (German: "der Beruf", masculine). However, existing research suggests that gender assignment may, in some cases, be etymologically motivated (Griebel, 2006), meaning that it derives from a word's historical form rather than its modern usage. For instance, *der Drink* is today associated with *das Getränk* (English: "the drink"), but is historically linked to *der Trunk* (English: "the drink") (Griebel, 2006).

Genus Principle

The genus principle (see Section 2.1.3), where certain groupings of German nouns receive a determiner based on their overarching genus can also be applied to English loanwords (Nyenhuis and Pittner, 2012; Vanhaelemeesch, 2020). This means that loanwords that are part of a specific semantic group will receive the grammatical gender that the overarching German noun receives. For example, for words related to winds (German: *der Wind*, masculine), such as *Tornado*, will receive the masculine gender according to the German equivalent.

2.3.5 Hierarchy in Gender Assignment for Loanwords

The hierarchy for gender assignment to English loanwords in German is in line with gender assignment to German words. They take the following order: biological gender > suffixes (morphological patterns) > translation > phonological patterns > *Gattungsprinzip* (overarching semantic topic) (Binanzer and Wecker, 2011) (Vanhaelemeesch, 2020) . It should be noted that the hierarchies describe the order in which gender is assigned, rather than the overall frequency of each gender. This is why the hierarchy and the frequency (see Section 2.3) differ.

2.4 Words Taking Multiple Genders

There are some words that allow multiple genders; this is called a *Genusschwankung* (Gender fluctuation). This is the case both for originally German and loanwords (Leibniz-Institut für Deutsche Sprache, 2017). These variations typically occur between masculine and neuter genders, while the feminine is only rarely an alternative. They arise for different reasons, such as variation between standard and regional varieties or ongoing processes of language change and borrowing (Eisenberg and Fuhrhop, 2016). In the context of loanwords, gender assignment may also follow competing principles, leading to multiple possible gender forms depending on the possible associations (Eisenberg and Fuhrhop, 2016). A few loanwords incorporated in the dataset used for this thesis take multiple genders. This will be highlighted in the dataset description and taken into account in the analysis of the results. It should be noted that for homonyms, there is mostly one binding gender depending on the reference (Leibniz-Institut für Deutsche Sprache, 2017). For example, *das Tau* (English: "the rope") is neuter, while *der Tau* (English: "the dew") is masculine.

2.5 Gender in Word Embeddings

Word embeddings are vector representations that are learned from a text-based corpus. They map words into a multidimensional vector space, where words with similar meanings appear closer together. In this way, they capture statistical regularities of language use within a given corpus. Natural Language Processing (NLP) tools are typically trained on data in their target language; however, there is a stronger incentive to develop English-language systems due to higher economic demand, greater availability of crowdworkers, and more funding (Yu et al., 2022).

Word embeddings do not only capture the similarities between words, but also encode a range of their semantic and syntactic properties learned from distributional patterns present in the training corpus. These embeddings can also reflect semantic information related to gender, as words associated with similar contexts tend to cluster together in the vector space. As a result, gendered associations present in the training data can be encoded in the representations.

Gender bias in word embeddings primarily reflects patterns present in the training data. In English systems, these biases typically emerge from social gender associations, for example, linking a doctor with male and a nurse with female referents. These associations arise from co-occurrence patterns with gendered pronouns in the underlying corpus (e.g., higher frequency of *she* with *nurse*) (Bordia and Bowman, 2019).

When discussing the gender biases in word embeddings, it is important to distinguish between the social biases mentioned above and biases introduced through grammatical gender. The gender marking in languages that have grammatical gender can affect the gender biases in word embeddings. In German, for example, there can be a lexical distinction between female and male individuals, such as a female doctor(*die Ärztin*) and a male doctor (*der Arzt*), or a female nurse (*die Krankenschwester*) and a male nurse (*der Krankenpfleger*). As a result, embeddings trained on such languages may reflect not only social gender associations but also grammatical gender marking, making it more difficult to isolate purely socially driven bias effects (Omrani Sabbaghi and Caliskan, 2022).

An example would be the grammatically feminine Spanish word *Fuerza* (English: "power"), which initially shows a strong association with the feminine grammatical gender, despite it being stereotypically masculine in meaning (Omrani Sabbaghi and Caliskan, 2022). This may also impact the measurements of social gender bias captured in an embedding space. Disentangling the signals coming from grammatical gender can help clarify whether an observed bias comes from the underlying social biases conveyed through the training data, or from the grammatical structure of a language (Omrani Sabbaghi and Caliskan, 2022).

The models used in the current experimental setup will have picked up some gender biases during pretraining. However, the primary focus of this thesis is not the analysis of bias within embeddings, but the prediction of grammatical gender for English loanwords in German. Therefore, any potential influence of social or grammatical bias on the model's internal representations is not examined in detail.

2.6 Transformer Models

Throughout the past nine years, since the publication of Attention is All You Need (Vaswani et al., 2017), transformer models have become a foundational architecture for NLP tasks, due to the efficiency with which they can model long-range dependencies using self-attention (Vaswani et al., 2017). In contrast to previous models, such as RNNs, transformers can process input in parallel, which allows for more efficient training (Reddy, 2024). Transformer models can be encoder and or decoder-based. This thesis will use bidirectional encoder models to see how transformers assign grammatical gender to loanwords.

2.6.1 BERT Architecture

BERT stands for bidirectional encoder representations from transformers (Devlin et al., 2018). BERT-style models are bidirectional encoder-only models that are pretrained on two tasks: masked language prediction (MLM) and next sentence prediction (NSP) (Devlin et al., 2018). MLM is the first step in pre-training, where random words (about 15% of the input text) are masked, and the model predicts them using the surrounding context (Chan et al., 2020). The second task is NSP, where the model predicts whether it is logical for sentence B to follow sentence A. Through this pretraining, BERT achieves promising results for many NLP tasks, and can be finetuned for many different NLP tasks, such as NER or negation detection (Rogers et al., 2020).

BERT takes text as an input, which is tokenised into word pieces/subtokens (Wu et al., 2016). These subwords are transformed into token-ids, as BERT only takes nu-

merical input. They are then converted into embedding vectors (Rogers et al., 2020). Each token is represented using three types of embeddings: a token embedding capturing its semantic meaning, a positional embedding (capturing word order) and a segment embedding (determining which sentence a token is part of) (Rogers et al., 2020). After the tokenisation, BERT combines the three embeddings into a fixed-length representation, with the [CLS] token being used for text classification and the [SEP] token to separate different input segments. This representation is then passed through the stacked transformer encoder layers, where each layer has multiple self-attention heads.

Furthermore, the subtokenisation of BERT allows it to deal with unknown words better. Other models have certain limitations that do not make them suitable for the task. For example, GPT models are unidirectional left context models (Ajayi, 2024) and predict the following word, whereas the right context can often be especially informative in German, as verbs often only come at the very end of a sentence (Eisenberg and Fuhrhop, 2016). The shortcomings of other models make BERT the ideal architecture for gender identification through determiner prediction.

Self-attention defines the ability of a model to attend to every other word in the same sentence or segment to understand the context in which a token is given. For each token, the self-attention head computes a query, key and value. Then, for each token, its query gets compared to the softmaxed values of all other tokens to see how relevant they are to each other. Tokens that are assigned high scores help the model understand the meaning of the current token, while low scores are assigned to less relevant words (Rogers et al., 2020). The queries, keys and values are used to produce a weighted representation of the sequence. These weighted representations or outputs of the self-attention heads are then combined to pass them through a fully connected layer, followed by a skip connection and layer normalisation (Rogers et al., 2020). All tokens are processed in parallel within one transformer layer, while each attention head computes its attention over a sequence of tokens in parallel to the other attention heads.

These embeddings are passed through the stacked encoder layers, also known as transformer blocks. There are two sizes of BERT models, which vary in the number of encoder layers: BERT base with 12 layers, and BERT large with 24 layers. Different layers of BERT are known to capture different information (Rogers et al., 2020). Previous research suggests that the lower layers tend to capture information related to word order, such as lexical features, POS tags, word order and morphological patterns. The middle layers (5-9) are said to capture syntactic structures, dependency relationships and subject-verb agreement and might therefore be the most significant for the current task. Furthermore, previous research suggests that the upper layers (9-12) tend to capture semantic information, sentence-level meaning and task-specific representations (after fine-tuning) (Rogers et al., 2020). The lower layers may therefore be important for the morphological patterns introduced in the dataset used for the current task, while the upper layers, capturing semantic information, may be of importance for loanword associations across languages and biological gender encoding.

During this pretraining process, the model is trained on large-scale unlabeled text corpora (Rogers et al., 2020). It allows a model to learn general linguistic representations from the text, including syntax, semantics and morphology. The objective is to create a general-purpose language model that captures the structure of one or multiple languages (Rogers et al., 2020).

2.6.2 Why BERT-style Models?

The task of predicting a loanword's gender via its nominative (or accusative) determiner treats determiner choice as a proxy for grammatical gender. This can be framed as a masked language modelling task, in which lexical properties of a token are inferred from its surrounding context. The predictions of a determiner rely on both the left and the right context. In German, preceding elements such as determiners and adjectives may provide cues about gender, while following words can also contain relevant information due to the Germans' flexible word order and frequent use of verb-final constructions. Consequently, successful gender prediction requires a model that can integrate information from the entire sentence rather than from a single directional context.

BERT-style models are particularly well-suited for this task because they have a bidirectional transformer architecture, allowing them to attend to both preceding and following tokens simultaneously. This provides an advantage over unidirectional models such as GPT, which only have access to left-side context during prediction. Though the subword tokenisation is not unique to BERT, it enables it to process previously unseen or infrequent loanwords by decomposing them into smaller meaningful units, which is especially beneficial when working with English-derived forms embedded in German text. Finally, BERT-style models are pretrained using a masked language modelling objective, making the prediction of a missing determiner closely aligned with the task on which the model was originally trained. Masked language modelling will be discussed in the following section.

2.7 Models

This section introduces the four BERT-style models used in the current experimental setup. One multilingual model (mBERT) and three monolingual German models are used for this task. All four models are used in the *cased* version. Each model is introduced briefly, followed by a comparison of the models.

2.7.1 mBERT

bert-base-multilingual-cased (mBERT) is a multilingual BERT-style model that has been pretrained on concatenated Wikipedia corpora from 104 languages (Libovický et al., 2019). It is a single model that can handle multiple languages and has been trained without language identifiers or explicit alignment signals (Wu and Dredze, 2020). Tokens from all languages are represented in the same embedding space, enabling cross-language transfer (Pires et al., 2019). It is able to perform tasks in a target language without specific training data and performs especially well on POS tagging and NER (Pires et al., 2019).

Due to its shared WordPiece vocabulary and joint multilingual pretraining, mBERT enables cross-lingual transfer, allowing representations learned from high-resource languages to generalise to low-resource languages (Pires et al., 2019). For example, some words, such as numbers or names, may be identical across many languages, and subword fragments such as *-ion*, originally derived from Latin, may frequently occur across many languages (Pires et al., 2019). The patterns, such as morphology and syntax, that mBERT learns during pretraining, may increase its performance on low-resource languages, especially those that are closely related within the same language family as a

high-resource language.

The multilingual pretraining of mBERT could be relevant for the prediction of grammatical gender in English loanwords embedded in German text. Since the model has been exposed to both English and German during pretraining, it may be able to use representations learned from both languages when processing anglicisms. This is especially relevant for loanwords that keep English morphological or written characteristics while appearing in German syntactic contexts. Through its shared embedding space, mBERT can potentially connect information about the English source word with cues provided by the surrounding German context, allowing it to make use of cross-linguistic knowledge when predicting the appropriate determiner. However, it should be noted that studies show that multilingual transformers, such as XLM-Roberta, are able to identify code-switches between German and English after being finetuned on the Denglish corpus (Sterner, 2024).

Furthermore, mBERT shows uneven performance across different languages, as high resource langauges dominate the embedding space (Wu and Dredze, 2020). As there is no explicit modelling of typological differences, the model can struggle with the rare constructions and fine-grained morphologies of some languages. German belongs to the high-resource languages, for which mBERT achieves better results in classification tasks; however, there are no exact percentages of the amount of German language data in mBERT pretraining data (Wu and Dredze, 2020).

This thesis includes a multilingual model to examine how cross-linguistic representations affect gender prediction for English loanwords in German sentences. The shared embedding space cross 104 languages enables transfer between languages without any explicit alignment, which could lead to both generalisation and interference from other languages. This allows the model to exploit similarities between German and related high-resource languages, such as English, but may also cause other languages to interfere. That mBERT has representations for the English and the German vocabulary, may support the alignment of words in the embedding space. For example, translation-based gender assignment may be facilitated due to the model's representations of the German equivalent.

2.7.2 German Models

germanBERT

Google's bert-base-german-cased (germanBERT) is a monolingual German version based on the BERT-base architecture introduced by Devlin et al. (2018). The model is pretrained on large-scale German corpora, including Wikipedia and OpenLegalData. It employs a WordPiece tokeniser and standard subword-level masking rather than whole-word masking. These differences will further be discussed in Section 2.7.3.

gBERT

GBERT is a monolingually trained bidirectional encoder trained only on German text. It was developed by DeepSet, trained specifically to outperform earlier German BERT variants. It was pretrained on a mixture of large German corpora, including German Wikipedia, common crawl, news crawl, and the EU-Bookshop corpus (Chan et al., 2020). The size of the training data was roughly 163GB of data, and 2.3 billion tokens (Chan et al., 2020).

GBERT shows a strong performance on German NLP benchmarks such as POS tagging and named entity recognition (Chan et al., 2020), suggesting that GBERT can capture German syntactic and morphological patterns. Its performance has improved compared to previous models, including germanBERT (Chan et al., 2020). It performs strongly on NER, POS tagging and German classification tasks. Furthermore, it has a more consistent representation of the German language than multilingual models (Velankar et al., 2022), due to reduced noise, including a strong encoding of agreement patterns and German's relatively complex morphology (which is especially important for gender marking). Unfortunately, due to the German-only training, the model has no cross-lingual knowledge and may not generalise well to English loanwords.

dbmdzBERT

bert-base-german-cased (dbmdzBERT) was developed by the Bavarian State Library NLP Group at the LMU (dbmdz, 2026). The model was pretrained on data from the same corpora as GBERT. However, there are differences in the filtering of the training data, as well as the training objectives, when compared to GBERT. These differences will be analysed in the following section. It is a strong general-purpose German language model, which is robust across different tasks due to its mixed range of training corpora and is seen as a stable baseline for grammatical tasks, such as gender agreement (Chan et al., 2020).

2.7.3 Model Comparison

This section will introduce the differences of the four models regarding training data, tokenisation and masking.

Training Data Differences

The first major difference in the training data is that mBERT has been trained on data from 104 languages, making it the biggest of the four models. A comparison of training data and model size can be seen in Table 2.4.

Model	Training Data Size (GB)	Parameters
mBERT	Unknown	177,853,440
germanBERT	12 GB (Google, 2019)	109,081,334
gBERT	163 GB (Chan et al., 2020)	109,927,680
dbmdzBERT	16 GB (dbmdz, 2026)	109,927,680

Table 2.4: Comparison of Models in Terms of Training Data Size and Parameter Count

The training data for gBERT and dbmdzBERT have a strong overlap. The corpora in the datasets they are trained on include: German Wikipedia, CommonCrawl, NewsCrawl, ParaCrawl, EU-Bookshop and OpenSubtitles (Chan et al., 2020). However, some of these dumps, such as the Wikipedia dump, differ every year. The data used was filtered for non-German documents (i.e. documents where the dominant language is not German). Furthermore, the pipelines filter for duplications of text, removing those that are almost the same, based on n-gram overlap similarity. Additionally, it is filtered for the quality of the texts, including linguistic structure (full sentences,

correct punctuation, grammatical structure), information density and structural completeness (Chan et al., 2020). Lastly, the size of the training data can vastly impact a model's performance, where an increase in training data correlates with better model performance (Chan et al., 2020).

GermanBERT is known to be trained on the following three corpora: German Wikipedia, OpenLegalData, and NewsCrawl. There is no further description of any preprocessing of the used data.

The training data used for mBERT consists of the entire Wikipedia dump, for each of the 104 languages that have the largest Wikipedias (Google, 2024). The data is not manually filtered beyond standard preprocessing, but an exponentially smoothed weighting scheme is applied to balance the distribution of languages.

Tokenisation

All four models are accompanied by their own tokeniser. I verified whether the three gold determiners used in the present task are further subtokenised by the models. The tokenisers of all models show that the determiners *der*, *die*, and *das* are represented as individual tokens rather than being split into subtokens.

Though the determiners are not split into subtokens, this may not be the case for all loanwords presented to the model in the current study, as they are likely to be unknown to the tokeniser and may therefore be split into more frequently occurring subwords. Depending on the model's vocabulary and tokenisation scheme, loanwords may be split into multiple subword units, which can vary across models. Such differences in tokenisation may influence how lexical and morphological information is represented, and could therefore affect the models' ability to consistently infer grammatical gender from the surrounding context. This will be investigated in the analysis.

Masking

BERT-style models can use one of two masking strategies: subword masking or whole-word masking (WWM). Subword masking masks part of a word, if it is split into subtokens. WWM masks all subtokens of a word. GermanBERT, dbmdzBERT and mBERT use subword tokenisation with token-level masked language modelling, while gBERT uses WWM (Chan et al., 2020).

The different masking strategies can influence how strongly models rely on morphological versus contextual cues when predicting the definite determiner. In subword masking, the models mask the individual WordPiece tokens independently (Devlin et al., 2018), which means that parts of a word remain visible during pretraining, allowing the model to learn relationships between subwords (such as suffixes) and grammatical gender.

2.8 Zero-Shot and Finetuning

The two settings in which the four models will be used in the current study are zero-shot and finetuning. This section provides an explanation of the two.

2.8.1 Zero-Shot Predicting

Zero-shot prediction is the application of a pretrained language model to a task without any additional training or information beyond what was learned during pretraining (Brown et al., 2020). There are no gradient or weight updates based on task-specific data, and the model's performance solely depends on how well pretraining matches task structure and the generalisation abilities of the model (Bergmann, 2024). For this thesis, a zero-shot model predicts the masked definite determiner for an English loanword embedded in a sentence, in the nominative or accusative case (e.g., "[MASK] Laptop ist neu." (English: "The Laptop is new.") based solely on the linguistic patterns acquired during pretraining, without having been explicitly trained on German grammatical gender prediction.

2.8.2 Fine-tuning

Fine-tuning refers to the adaptation of a pretrained language model to a specific task using labelled, task-specific data. During this process, the weights are updated through supervised training using a task-specific head, such as a classification layer or a sequence labelling head (Rogers et al., 2020).

In this thesis, four pretrained models will be finetuned on a manually created dataset where English loanwords are incorporated into grammatical German sentences using the nominative or the accusative case. The determiner is the masked token and will be predicted as a proxy for grammatical gender.

Unlike continual pretraining, which further trains the model on large amounts of unlabelled domain-specific data, fine-tuning trains the model for a specific task (Rogers et al., 2020). As a result, fine-tuning shifts the model's representations towards task-relevant features, while simultaneously making use of patterns acquired during pretraining.

2.9 Previous Studies on Grammatical Gender with MLMs

This section reviews two studies that have previously investigated the prediction and representation of grammatical gender in neural language models. The earlier study focuses on gender prediction in Swedish using embeddings and a standard feed-forward neural network (Allassonnière-Tang and Basirat, 2020). The more recent study focuses on gender and number in English, German and Spanish (Janzso, 2021).

2.9.1 Predicting Grammatical Gender in Swedish

The distinction between the three grammatical genders found in older Germanic languages has merged into a binary distinction in Swedish. The masculine and feminine grammatical genders have merged into one common uter (utrum) gender. Nouns can therefore be divided into the utrum category (*en* words) or the neuter category (*ett*) words. For example; *en man* (English: "a man"), *en kvinna* (English: "a woman") and *ett barn* (English: "a child"). Gender in Swedish is marked through an indefinite determiner or through definite suffixes *-en* for utrum and *-et* for neuter) (Borg and Alemán Bañón, 2026).

In Swedish, the utrum gender is more dominant, making up around 70% of nouns. The distribution of the two genders in the training, development and test data is kept

similar to the original distribution in Swedish. Therefore, the baseline against which performance is measured it 70% (Allassonnière-Tang and Basirat, 2020).

Work prior to this study investigates gender assignment in Swedish using a basic feed-forward neural network in combination with word embeddings (Allassonnière-Tang and Basirat, 2020). The word embeddings are obtained from a large raw corpus, and the feed-forward neural network maps each noun's vector either to the utrum or the neuter gender. or The authors treat gender as a lexical property that can be inferred from the surrounding context. According to them, the assignment of gender to a noun can be done by three approaches: selection by chance, the noun's internal form (morphological or phonological patterns) or an analysis of the surrounding context (Allassonnière-Tang and Basirat, 2020).

Their 50-dimensional word-embedding in combination with the basic neural classifier achieved 92% accuracy, where utrum nouns were classified more reliably (Accuracy: 95.40%) than neuter nouns (Accuracy: 84.16%). Their results showed that the best results came from an asymmetric-backwards window and that larger windows introduced noise and lowered performance, especially for neuter nouns. The error analysis highlights that the three main error sources are noise, nouns isolated from context and the polysemy of nouns that are linked to more than one gender depending on their sense (Allassonnière-Tang and Basirat, 2020).

2.9.2 Grammatical Gender in BERT-Style Models

Whether BERT can distinguish between grammatical number and gender has been investigated by (Janzso, 2021). She investigated the abilities of four BERT models (BERT-base-uncased, Bert-base-multilingual-cased, bert-base-german-uncased and bert-base-spanish-wwm-uncased) to distinguish between number and gender across three languages: English, German, and Spanish. However, the strongest conclusions are drawn on German, as English does not have grammatical gender, and the Spanish data was too sparse to draw quantitative conclusions.

The paper focuses on whether BERT-style models are able to distinguish the gender of nouns that can occur with two genders depending on their sense. For example, *die Kiefer* (feminine), translates to *the pine-tree*, while *der Kiefer* (masculine) translates to *the jaw* (Janzso, 2021). The findings show that the models germanBERT (BERT-base-German-uncased) and mBERT (BERT-base-multilingual-uncased) contextual embedding only separate ambiguous gender pairs slightly from non-ambiguous pairs. The cues that the four models use to identify gender are the determiner and the gender-marked adjectives.

The monolingual BERT models outperform the multilingual ones when it comes to both gender and number agreement. According to Janszo (Janzso, 2021), the strongest cues for gender assignment are morphological (suffixes), adjective agreement and the determiner. BERT can handle number disambiguation better than gender disambiguation.

To my knowledge, no prior work has addressed grammatical gender assignment in transformer-based models for English loanwords have been found. This highlights a gap in the current literature on grammatical gender assignment in transformer-based models, particularly for English loanwords in German.

Chapter 3

Dataset Description

3.1 General Description

The dataset used for this thesis was created manually and is based on the patterns by which grammatical gender is assigned to English loanwords that are introduced in 2.3. It consists of 989 annotated German sentences containing an English loanword in the nominative or the accusative case (i.e., with one of the following three determiners, indicating its gender: *der*, *die* or *das*). Masculine words in the accusative case are not part of the dataset, therefor the determiner *der* is not listed. The majority of sentences use the loanword in the nominative case. Loanwords used in the accusative case have either feminine or neuter grammatical gender. The determiners used for the feminine and neuter loanwords in the accusative case are the same as those in the nominative case. Therefore, no additional adjustments are required for the analysis of alternative predictions, and these determiners can be used as a proxy for grammatical gender, just as nominative determiners are. This will be described further in the results section (5.1).

3.2 Categories and Subcategories

The sentences in the dataset are organised into four primary categories, based on the previous literature discussed in Section 2.3. An additional category is added as a control condition. Each category is further divided into subcategories.

The four main categories are: morphological mapping, biological gender, translation, and phonological patterns. The morphological pattern category comprises loan nouns with their translated derivational suffixes, which correlate with a specific gender assignment. The biological gender category focuses on the biological sex of the referent, with a distinction between nouns referring to a male or a female individual. The translational category includes loanwords whose translation corresponds to a specific gender. Lastly, the phonological category assigns gender based on the phonological shape principles introduced in section 2.3.3, including monosyllabicity and final plosives.

The subcategories into which each category can be divided are visualised in Table 3.1. The morphological mapping class has the five subclasses, one per equivalent suffix: *-er*, *-ing*, *-ion*, *-ity*, and *-ness*. The biological gender class has one subclass per biological gender (male and female). The translation class has three subclasses (one per grammatical gender). Lastly, the phonological class has two subclasses, as introduced above. The subclasses contain different numbers of examples, as the distribution

of classes across the dataset was designed to be relatively balanced. In addition, an effort was made to reduce differences between the number of examples per grammatical gender; however, this could not be fully achieved due to a limited number of loanwords found for certain subcategories.

Category	Subcategory	Grammatical Gender	Number	Example	English Translation
Morphological Mapping	-er	Masculine	60	Der Manager arbeitet viel.	The manager works a lot.
Morphological Mapping	-ing	Neuter	92	Das Meeting beginnt jetzt.	The meeting starts now.
Morphological Mapping	-ion	Feminine	53	Die Reaktion war positiv.	The reaction was positive.
Morphological Mapping	-ity	Feminine	51	Die Aktivität macht Spaß.	The activity is fun.
Morphological Mapping	-ness	Feminine	40	Die Fitness ist wichtig.	Fitness is important.
Biological Gender	Feminine	Feminine	94	Die Queen sah sehr gut aus.	The queen looked very good.
Biological Gender	Masculine	Masculine	95	Der Boss verhandelt.	The boss negotiates.
Translation	Masculine	Masculine	84	Der Job ist stressig (Beruf).	The job is stressful (occupation).
Translation	Feminine	Feminine	80	Die Party war lang (Feier).	The party was long (celebration).
Translation	Neuter	Neuter	80	Das Team gewinnt das Spiel.	The team wins the game.
Phonological	Monosyllabic	Masculine	90	Der Deal ist gut.	The deal is good.
Phonological	Final Plosive	Masculine	90	Der Trip war kurz.	The trip was short.
Plurals	-	Feminine	80	Die Teller sind leer.	The plates are empty.
				Total number of examples	**989**

Table 3.1: Dataset Class Distribution with Number and Examples

The total number of example sentences is 989, with 961 unique loanwords, as some were repeated across classes. For example, the loanword *Manager* occurred under the morphological mapping subclass *-er* and under the male biological gender subclass. For which classes loanwords were repeated is shown via a confusion matrix in appendix A.2.

As all plural nouns in German take the determiner *die* in the nominative or accusative case, regardless of their singular grammatical gender, the plural determiner cannot serve as a proxy for gender assignment (see Section 2.3.2). A category for plural loanwords was therefore included as a control condition rather than a gender category. The correct prediction of the determiner *die* for plural loanwords would indicate that models are sensitive to number agreement cues in the surrounding sentence, such as plural verb inflexion or the English plural marker *-s*. As grammatical gender is not expected to influence determiner selection in these instances, little variation across models and settings is expected.

To probe the models' understanding of gender for the loanwords included in the plural condition, an additional test set containing the singular forms of these nouns was included as part of the analysis (see section 6.5). Other tests are described in the following section.

3.2.1 Additional Test Sets

In addition to the main dataset, two other small datasets were created. These datasets were used for further analysis of the results. The first, smaller test set, looks at the model's reliance on context by using nouns referring to a biological gender and using them with context words that are typically associated with the opposite gender (see example 3). This test includes five examples of loanwords referring to male individuals and five examples referring to female individuals.

(3) Der Stalker trägt ein schönes Kleid
DEF.M.SG stalker.M carry.3SG.PRS INDEF.N.SG beautiful dress
'The stalker is wearing a beautiful dress.'

Furthermore, a semantic test set based on the *Gattungsprinzip* (genus principle) was constructed to evaluate model generalisation beyond the rule-based categories used to create the main dataset and during training. Unlike the morphological, phonological, and translation-based categories, the *Gattungsprinzip* relies on broader conceptual associations rather than productive formal cues, making it a suitable test case to probe whether the models can extend the learned gender assignment patterns to more abstractly motivated instances. This test set contains one subcategory per grammatical gender. For the masculine gender, loanwords are associated with the overarching terms of *wind* and *precipitation*, following the overarching German nouns *der Wind* and *der Niederschlag*. These are comprised under the subcategory *Natural Elements*. For the neuter gender, loanwords belong to the domain of technology, are grouped under the overarching noun *das Gerät* (English: "the device"). For the feminine gender, all loanwords are related to events, grouped under the overarching term *die Veranstaltung* (English: "the event"). This testset contains 11 examples for Natural Elements (*masculine*), 17 examples for technology (*neuter*) and 24 examples for events (*feminine*). The loanwords used for this test set should take the gender of the overarching category.

Because these associations are less binding than the patterns included during training, this test set specifically targets a model's ability to generalise to loanwords of

more loosely defined semantic categories that were not seen during finetuning. A few loanwords included in this test set may also have characteristics associated with other subclasses introduced above; this does not lead to conflicting gender assignments, as the expected gender remains consistent across the different assignment principles.

3.3 Dataset Creation

The sentences made for this data set were created using an Excel file during the creation process. Each instance in the dataset includes the following coloumns: SentenceID, sentence, category, subcategory, gold gender, dictionary label, and loanword (see Figure 3.1). The gold gender is defined as the gender assigned according to the subclass to which the loanword belongs, whereas the dictionary label reflects the grammatical gender assigned in the dictionary. In most cases, the dictionary label and the gold label (based on subcategory) are identical; however, some loanwords are assigned multiple genders in the dictionary (see Section 2.4), in which cases, all possible genders are included under the annotation *dictionary label* and will be taken into account during the analysis of the results.

To create the sentences, I created a list of anglicisms I frequently use in my daily life. Each loanword was looked up in the standard German dictionary *Duden*. The dictionary provided information on the grammatical gender and determiners associated with the noun for all grammatical cases, as well as the word's origin.

For each word, it was ensured that the word's origin was English. An exception to this is loanwords ending in *-ion*. The loanwords ending in *ion* that were included in the current dataset are originally Latin. However, these instances are treated as borrowings from English, as they enter German through English rather than directly from Latin. This was verified based on the information provided by **Duden** for every loanword of this kind included.

The determiner gold labels for each loanword determiner are based on the gender assigned by *Duden* that corresponds to the gender of the subclass.

Sentence ID	Satz	Category	Subcategory	Gender	Duden	Loanword
1	Der Manager hat eine Entscheidung getroffen.	Morphological Mapping	-er	Masculine	M	Manager
2	Der Influencer hat ein Bild veröfentlicht.	Morphological Mapping	-er	Masculine	M	Influencer
3	Der Follower hat den Beitrag kommentiert.	Morphological Mapping	-er	Masculine	M	Follower
4	Der Browser ist schnell.	Morphological Mapping	-er	Masculine	M	Browser
5	Der Router ist kaputt.	Morphological Mapping	-er	Masculine	M	Router
6	Der Sleeper ist super bequem.	Morphological Mapping	-er	Masculine	M	Sleeper
7	Der Booster hat mir geholfen.	Morphological Mapping	-er	Masculine	M	Booster
8	Der Bestseller wurde tausende Male verkauft.	Morphological Mapping	-er	Masculine	M	Bestseller
9	Der Oldtimer wurde neu lackiert.	Morphological Mapping	-er	Masculine	M	Oldtimer
10	Der Camper fährt auf der linken Spur.	Morphological Mapping	-er	Masculine	M	Camper
11	Gestern ist der Computer abgestürzt.	Morphological Mapping	-er	Masculine	M	Computer
12	Der Hacker ist ins System eingedrungen.	Morphological Mapping	-er	Masculine	M	Hacker
13	Nach dem Konzert hat der Rapper Autogramme gegeben.	Morphological Mapping	-er	Masculine	M	Rapper
14	Dem Jungen hat der Cheeseburger geschmeckt.	Morphological Mapping	-er	Masculine	M	Cheeseburger
15	Mir ist gestern der Toaster kaputt gegangen.	Morphological Mapping	-er	Masculine	M	Toaster

Figure 3.1: Raw Dataset Structure

During the search process, for one loanword, more were found in the **Duden**. After concluding my search through the dictionary based on the personal list, I expanded my search to *the Anglizismen-Index*. This book stores frequently used anglicisms in German (Verein Deutsche Sprache and Verein Muttersprache Österreich, 2023). All loanwords taken from this index were searched in the *Duden* to obtain their corresponding gender and added to the best-fitting category. For about 10% of the loanwords included in the data set, there was no entry in the *Duden* that specified their gender. In these cases, the rules described in the previous literature (based on the ranking) were used

to classify loanwords into the five categories, when in line with the judgment of an L1 speaker. To verify the grammatical correctness of the examples and the usage of the loanwords in general, as well as their correct usage in the context of the created sentences, a German studies teacher reviewed them.

All sentences were created manually without the help of AI. For most examples, the noun and its determiner often occur towards the beginning of a sentence construction. To ensure syntactic variation, questions and passive constructions were included. For most sentences, the determiner directly precedes the loanword (see Example 4). However, some constructions include adjectives between the determiner and the loanword (Example 5). Lastly, several sentences use pronouns that agree with the gender of the agents and patients in the construction when they are human (Example 6). Efforts were made to minimise potential lexical cues, such as the use of gender-marked adjectives or pronouns for loanwords.

(4) Der Ball ist rund
the ball be.3SG round
'The ball is round.'

(5) Der blaue Ball ist rund
the blue ball be.3SG round
'The blue ball is round.'

(6) Der Ball, der ihr gehört,
the.DEF.M.NOM.SG ball.M.SG REL.NOM.SG 3SG.F.DAT belong.3SG
ist rund
be.3SG round
'The ball that belongs to her is round.'

The sentences in the dataset are in the nominative case (for all three genders) or the accusative case (for feminine and neuter loanwords). As a result, the gold labels are always *der*, *die* or *das* (see Table 2.1). These forms can also be realised by alternative determiners, including indefinite determiners as well as possessive, demonstrative, or distributive pronouns. This is further discussed in Section 5.1 and taken into account in the evaluation and analysis of the results.

3.4 Formatting

For easier usage and masking during the coding process, a CoNLL-style file was created, with the sentence-level information added for each instance, as well as token-level information, indicating which token is the loanword and which token is the determiner that should be masked. Figure 3.2 shows an example sentence in this CoNLL format. To convert the Excel table format to CoNLL-style files, I manually created a sample in the desired CoNLL format for two rows in the Excel file and prompted ChatGPT to apply it to all other examples. The linguistic annotations, of which word is the loanword and which the corresponding determiner, were added to the CoNLL style format manually. No linguistic decisions were AI-generated. The prompt and the first couple of responses can be seen in the Appendix A.1.

```
#sentence_id: 33
#category: morphological_mapping
#subcategory: -er
#gender: masculine
#duden: M
#noun: Camcorder
#example_number: 33
#text: Der Camcorder ist noch wie neu.
1       Der     -       D
2       Camcorder       N       -
3       ist     -       -
4       noch    -       -
5       wie     -       -
6       neu     -       -
7       .       -       -
```

Figure 3.2: CoNLL-Style Dataset

3.5 Intended Use

This dataset was created for this thesis and is intended to be used for masked token prediction of definite determiners for English loanwords using BERT. However, it may also be used to train other machine learning algorithms and evaluate their ability to predict grammatical gender, number or determiners correctly. Furthermore, it may be used to investigate questions related to the use of anglicisms in German.

3.6 Dataset Split

When training machine learning models, it is common to split a dataset into a training, development, and test set. The training set is used to train the model, and the development set is used to test its performance and tune the hyperparameters for better performance. The model is then evaluated using the test set. The dataset used for this thesis is relatively small. It is split into a training, development, and test set with a split of 2:1:1, respectively. This is done using a Python script to ensure reproducibility and to enable the creation of consistent splits for potential cross-validation experiments. As an equal split is trying to be maintained, one sentence of the plural instances included in the original dataset did not make it into the training, development, and test set.

The data is split to maintain the original proportion of examples from all subcategories. The complete dataset contained a total of 13 subcategories with the division shown in Table 3.2.

The gender distribution can be seen in Table 3.3. The number of examples included in this table is lower than the total number of examples, as all instances belonging to the plural category do not allow for a distinction of grammatical gender.

The first step is to calculate the total number of instances in the data set by summing the size of all groups. The relative proportion of each subcategory within the full dataset is calculated. This proportion is used to determine the number of examples of each subcategory that should be assigned in each split. This is done by multiplying the

Category	Subcategory	Gender	Train	Dev	Test
Morphological Mapping	-er	Masculine	30	15	15
Morphological Mapping	-ing	Neuter	46	23	23
Morphological Mapping	-ity	Feminine	25	13	13
Morphological Mapping	-ion	Feminine	27	13	13
Morphological Mapping	-ness	Feminine	20	10	10
Biological Gender	Female	Feminine	46	24	24
Biological Gender	Male	Masculine	47	24	24
Translation	Masculine	Masculine	42	21	21
Translation	Feminine	Feminine	40	20	20
Translation	Neuter	Neuter	40	20	20
Phonological Patterns	Monosyllabic	Masculine	44	23	23
Phonological Patterns	Final Plosive	Masculine	44	23	23
Plurals	-	-	39	20	20
Total	-	-	490	249	249

Table 3.2: Distribution of Subcategories Across Train, Development, and Test Splits

Gender	Train	Dev	Test	Total
Masculine	207	106	106	419
Feminine	158	80	80	318
Neuter	86	43	43	172
Total	451	229	229	909

Table 3.3: Grammatical Gender Distribution Across Dataset

proportion by the desired total size of each split and rounding the result to the nearest integer. After this allocation, a check is done to see if the sum of the assigned instances across the three splits exceeds the available number of examples in that group (this may occur due to rounding). If there is a surplus of examples assigned to a category, then the excess is computed, and the number of examples of that category allocated to the training set is reduced accordingly. The final result stores the subcategories and the number of examples assigned to each subcategory for the three different data splits as shown in Table 3.2.

Computing the stratification this way can reduce class imbalance risks, ensuring that there is not a single category that strongly dominates any of the data splits. It is particularly important because some classes are rarer than others and could therefore be underrepresented or entirely absent from one of the splits under random splitting, without stratification.

The dataset was randomly shuffled to remove any potential ordering effects. It was then split into three separate files (training, development, and test). This ensures that each dataset contains an equal proportion of examples from each category while preventing systematic biases that may arise from the original ordering of the data.

3.7 Limitations and Biases

There are several limitations that come with manually creating a dataset. The first limitation is the selection bias that comes with manually constructing a dataset. The loanwords included in the dataset were partially chosen based on familiarity or frequency of use. Though the loanwords stemming from the index of anglicisms are not affected by this, this may still have led to an overrepresentation of commonly used anglicisms. Furthermore, the dataset is limited to 989 sample sentences (with 961 unique loanwords), which reduces the overall coverage, as well as the likelihood of very rare loanwords being included. The dataset may therefore not fully reflect the natural distribution of anglicisms in German. Lastly, for the loanwords that overlap across categories, it cannot be clearly separated, which cues the model relied on.

Furthermore, there may be an annotation bias. The categories and subcategories into which loanwords are divided are theory-driven. This may impose a predefined linguistic structure that may not always reflect language use. The gender labels were primarily based on the *Duden* dictionary entries. This may introduce a bias towards a normative standardised language use. For loanwords that did not occur in the *Duden*, the gender was assigned based on these rules. The gender assignment in the *Duden* however, is corpus and frequency-based (Wolf, 2021).

Though a manually constructed dataset allows for controlled conditions, the sentences into which the loanwords are embedded may have caused reduced linguistic variation. An effort was made to include variation, such as questions, passive constructions, and adjectives. However, these constructions may introduce biases or hints to the model, such as adjectives inflected for a specific gender.

Another limitation is that care was taken to minimise the introduction of additional gender cues in the sentences. In particular, lexical and contextual cues that could explicitly signal biological gender were reduced where possible, for example, by avoiding gender-marked adjectives or referential pronouns. This may reduce how natural the sentences are.

Despite these limitations, the dataset was created in good faith with the goal of providing a controlled experimental environment for investigating grammatical gender assignment in English loanwords in German and allows for a systematic evaluation of masked language models on this task.

3.8 Ethical Considerations

The dataset does not contain any personal or sensitive information. All examples were manually constructed for the purpose of this thesis. No participant data or copyrighted corpora were used to create the dataset. However, there are some ethical considerations in relation to possible gender biases mentioned in Section 2.5.

Another ethical consideration is the use of a single gold standard for grammatical gender. While the gold gender labels used for this thesis were selected based on the literature and existing dictionaries, variation exists in the gender assignment of some English loanwords. As a result, speakers who use a different gender for a particular loanword may not be represented by the gold labels. The results should therefore not be interpreted as prescribing one correct grammatical gender assignment, but rather as measuring model predictions against the grammatical gender labels selected for this study.

Chapter 4

Methodology

This thesis aims to compare the abilities of multiple BERT models in predicting the determiners for English loanwords embedded into German sentences using the determiner as a proxy for grammatical gender.

This Chapter outlines the methodology used to evaluate masked token prediction with BERT models in both zero-shot and fine-tuned settings. It first describes the zero-shot and fine-tuning pipelines, as well as the evaluation and analysis procedures.

Predictions made in the zero-shot setting rely on patterns models have learned during pretraining, without updating their weights. In contrast finetuned predictions are produced by models that were additionally trained on the training split of a manually created dataset.

Four models were used for each of the two settings, resulting in eight evaluation setups. The models evaluated are introduced in Section 2.7, which also discusses their pretraining.

The link to all code used to train models, predict and analyse results is provided in the appendix (A.5).

4.1 Zero-Shot Approach

In the zero-shot setting, four BERT models make predictions for the masked determiner of English loanwords using the test set. The task is framed as a masked language modelling problem, where the determiner is replaced by the special [MASK] token. The model has to predict the missing token based on the contextual information learned during pretraining.

4.1.1 Preprocessing and Masking

In the zero-shot setting, the models were evaluated both on the entire dataset (all 989 sentences) and the test set (for comparison to models in a finetuned setting). Each input file has a CoNLL-like structure, as described in the dataset transformation section.

Each sentence contains annotations identifying the English loanword and its corresponding determiner. The determiner is replaced with the [MASK] token (see example 7), and the model must predict the missing determiner from the context. This ensures that predictions are conditioned on the surrounding linguistic context rather than just the loanword itself.

(7) Ich habe [MASK] Tür geöffnet
I have [MASK] door open.PTCP
'I have opened the door.'

4.1.2 Pipeline

For prediction, the HuggingFace "fill-mask" pipeline is used. This provides a standardised interface for masked language modelling, where sentences containing a [MASK] token are directly processed by pretrained BERT models. The pipeline handles tokenisation and inference using a model's corresponding tokeniser.

A key step in this pipeline is the subtokenisation, which splits input tokens into subword units to handle unseen or rare words. Example 8 shows the subtokenised version of example 7. The resulting tokenised sequence is then used by the model to compute contextual representations. The model produces a contextual embedding for the [MASK] position, which is used to derive a probability distribution over the vocabulary. From this distribution, the most likely token candidates are ranked. The top 10 predictions per instance are retained to allow both a quantitative evaluation (top-1 accuracy) and a qualitative analysis of alternative predictions. A batch size of 16 is used during inference to balance computational efficiency and memory constraints.

(8) [CLS] Ich hab##e [MASK] Tür ge##öffnet . [SEP]
I have [MASK] door open.PTCP
"I have opened the door."

For evaluation, only the highest-ranked prediction for each masked token is used, as this allows for comparison with the gold determiner. The top-10 predictions are additionally stored for further qualitative and error analysis, as evaluation based solely on the highest-ranked prediction may be overly strict in some cases. The performance is then evaluated using confusion matrices and classification reports across determiner classes, grammatical categories, and subcategories. This enables a detailed assessment of model behaviour across different linguistic phenomena.

4.2 Finetuning

To compare the zero-shot predictions with those of a fine-tuned model, the four previously mentioned models were fine-tuned on the task of determiner prediction for English loanwords in German sentences, used as a proxy for grammatical gender. This was done using the dataset splits created earlier, where the training and development sets were used for training and optimisation, and the test set was reserved for final evaluation. This allows for a direct comparison between pretrained (zero-shot) and task-adapted (finetuned) model performance under identical evaluation conditions.

As the fine-tuning setup is slightly more complex than the zero-shot setting, each model was fine-tuned separately using identical experimental configurations (see Appendix A.4). All models share the same training pipeline and hyperparameters to ensure comparability across architectures.

After loading the datasets, the sentences are tokenised using the tokeniser corresponding to each model. The tokeniser converts input tokens into subword units and corresponding token IDs using each model's corresponding tokeniser. This ensures compatibility with the pretrained vocabulary of each model. Furthermore, padding

and truncation are applied to ensure that the input lengths for each instance are kept the same and to avoid sequences exceeding the maximum model input length. This standardisation is required for efficient batch processing during training.

For this setup, no hyperparameter tuning was performed. The models are trained on the training set and evaluated on the development set after each epoch. Each model was fine-tuned for three epochs. The models were finetuned using the same core hyperparameter configuration (see Appendix A.4). However, there are differences for the tokenisers, as each model uses its associated pretrained tokeniser. Because the primary objective of this thesis is not to identify the optimal hyperparameter settings for each model, but rather to compare performance in zero-shot and fine-tuned settings, using a shared training configuration without hyperparameter tuning allows the analysis to focus on the effects of fine-tuning rather than on model-specific optimisation strategies. Though hyperparameter tuning may slightly improve the performance of the models, this would create a new experimental variable for comparison (zero-shot vs finetuned vs hyperparameter tuned), which would exceed the scope of this thesis.

4.2.1 Label Alignment

During finetuning, the gold labels must be aligned with the tokenised input sequence due to subtokenisation. Because tokenisation splits words into subword units, the original token-level labels do not directly correspond to model inputs.

Instead of directly aligning all token labels, a new label sequence is created with the same length as the tokenised input. All subtoken labels are initially assigned the value -100, indicating that they should be ignored during loss computation and evaluation, following the standard HuggingFace convention for ignored tokens during loss computation. This is done to ensure that only the relevant prediction targets contribute to the training signal. Figure 4.1 shows an example sentence, the padded tokenised input the models receives and its aligned labels.

```
Sentence:
Ich habe [MASK] Brightness des Displays hochgestellt .

Tokenised Input
['[CLS]', 'Ich', 'habe', '[MASK]', 'Brig', '##ht', '##ness', 'des', 'Display', '##s', 'hoch', '##gestellt', '.', '[SEP]', '[PAD]', '[PA
D]', '[PAD]', '[PAD]', '[PAD]', '[PAD]', '[PAD]', '[PAD]', '[PAD]', '[PAD]', '[PAD]']

Aligned Lables
[-100, -100, -100, 128, -100, -100, -100, -100, -100, -100, -100, -100, -100, -100, -100, -100, -100, -100, -100, -100, -100, -100, -10
0, -100, -100]
```

Figure 4.1: Example Sentence, Tokenisation and Label Alignment

The gold determiner label is converted into its corresponding token ID and assigned as target output for the masked position. All other output is ignored. This design ensures that the model learns to predict only the masked determiner while still receiving the full sentence context as input.

4.2.2 Huggingface Pipeline and Datasets

The preprocessed data is converted into HuggingFace Dataset objects to enable compatibility with the trainer. This allows for batching, shuffling, and usage of the HuggingFace training pipeline. The resulting dataset structure is used for fine-tuning via the HuggingFace Trainer, ensuring consistency in input formatting across experimental conditions.

The fill mask pipeline takes a sentence containing a [MASK] token as input, tokenises it using the model-specific tokeniser, and passes it through a pretrained transformer model. The model then produces a contextualised representation of the masked token and uses it to compute a probability distribution over the vocabulary, returning the highest-ranked candidate tokens for the masked position, as well as their confidence scores.

4.2.3 SEEDs

When training a model, there are multiple random components, such as data shuffling and parameter initialisation. Because of these random differences, running the same model with the same settings can still lead to slight performance differences. Setting a random seed fixes the random number generator and, therefore, makes the individual runs reproducible. Evaluating the same model across multiple random seeds provides a more robust estimate of its performance. This allows the stability of the model to be assessed, rather than relying on a single run. Therefore, each model was trained and evaluated on the development set using four different random seeds.

SEED Testing

The following four random seeds are used for each model: 33, 42, 123, 3210. This is done to test how reliable the results are and creates slightly different runs each time, due to differences in data shuffling, model initialisation and the training process. It helps to identify whether the performance differences are due to random initialisation. The results from the different seeds are combined to calculate the average performance and the deviation thereof.

Fixed SEED

For the detailed analysis of the model's results, one fixed seed of 33 is used. A fixed seed is used because qualitative analysis requires consistent and reproducible model outputs. Using multiple different seeds would make it difficult to compare individual predictions, as outputs would vary across runs. The fixed run is used to examine results, analyse error patterns and inspect the top-k predictions.

4.2.4 Evaluation Metrics

The results of each model's predictions are presented in a three-way split, classifying predictions as correct, underspecified, or incorrect. The differences between these three groups are explained in Section 5.1. Results are reported as the proportion of predictions that fall into each category for both the zero-shot and fine-tuned settings.

This evaluation setup was chosen because not all grammatical predictions allow the gender of the loanword to be inferred. A binary distinction between correct and incorrect predictions would therefore fail to capture cases in which predictions form a grammatical sentence without clearly indicating a grammatical gender.

Unless otherwise stated, all results are based on the top prediction for each masked token. During the analysis, more than just the top prediction may be taken into account to investigate alternative gender assignments and model confidence. All predictions are stored in two formats, as described in the following section.

4.3 Storing Results

The results of each model's output, both in a zero-shot and fine-tuning setup, are stored in two different file formats. The first file format contains one row per sentence, including the sentence, the sentence ID, the category and subcategory, the loanword, the gold determiner and the top 10 predictions in order. The second file contains the same metadata, and additionally the tokenisation of the loanword, and the probability logits and softmaxed confidence scores for each of the top-k predictions. This separation allows for both high-level performance evaluation and a more fine-grained inspection of the models' uncertainties and ranking behaviours.

4.4 Additional Tests

All four models generated determiner predictions for both the context-based and semantic-grouping test sets (see Section 6.5), before and after fine-tuning. These tests were not part of the training data. In the zero-shot setting, the prediction procedure was identical to that used for the main test set. Since the models were not fine-tuned on these additional datasets, the fine-tuned versions of all four models were simply loaded and used to make predictions on these two test sets.

4.5 Analysis

The analysis of the results is done through scripts and therefore counts towards the methodological aspect of this thesis. The analysis can be split into two sections: an overview of alternative predictions and an error analysis. The aim of the thesis is to investigate the behaviour of the models when predicting determiners for English loanwords embedded into German sentences.

There are alternative predictions to the direct nominative and accusative determiners that can still form a grammatical sentence and provide gender insights, including the following: indefinite determiners (*ein* and *eine*), adjectives (correctly inflected for the noun), possessive pronouns (*mein*, *meine*, etc.) and demonstrative pronouns (*diese*, *dieses*). This is further described in Section 5.1.

Taking the alternative predictions into account, a standard error analysis is performed on all predictions that deviate from the gold and are not considered valid alternative predictions. These will be analysed with respect to patterns by class and subclass, loanword and tokenisation length, loanword frequency, and confidence intervals.

Class and Subclass patterns were investigated based on the class divisions in the test set. To determine patterns based on the length of loanwords, the number of characters a loanword is composed of is used as a unit of measurement. For the tokenisation length, the tokenisation patterns of each loanword by each model were stored when storing the predictions. Though each model has its corresponding tokeniser, tokenisation was consistent across models. The number of subtokens a word is composed of is used as a unit of measurement.

The pretraining data of germanBERT and dbmdzBERT contained the NewsCrawl. NewsCrawl offers frequency lists of all words in the training data. These frequency lists were used to determine which loanwords occurred more frequently during pretraining.

The rank based on this frequency list is then used to determine if the gender of more frequent loanwords has a higher chance of being predicted correctly.

4.6 Survey

To complement the analysis of Loanwords accepting multiple genders, German L1 speakers were invited to complete a Qualtrics Survey. The survey included all 17 loanwords from the original dataset for which the *Duden* allows multiple genders. Participants were presented with the loanword and the sentence in which it occurs in the dataset, with the determiner replaced by [MASK], giving them the same context as the BERT models. The answer options were presented in a multiple-choice format, forcing participants to choose one gender per loanword, to see their preference. An example extract from this survey is presented in the Appendix A.7. The order of the questions was kept constant across participants.

While this forced-choice design may not fully reflect the natural variation in German L1 speakers' judgments, it was done to ensure a directly comparable evaluation against the top model predictions, in which the highest-ranked prediction is used as the primary output for comparison.

A total of 10 participants completed the survey. Participation was voluntary and anonymous. Participants are German L1 speakers who have completed their secondary education in English. This allows them to understand the English vocabulary, while still having a stronger German background used to generalise German grammar rules over loanwords.

The results obtained from this survey will be compared with model predictions in the analysis (Section 6.3.2).

Chapter 5

Results

This section presents the results of all four models in predicting the determiner for English loanwords embedded into German sentences, in both the zero-shot and the finetuned setting. The determiner is used as a proxy for grammatical gender.

The results are divided into a three-way split: correct, underspecified and incorrect predictions. Correct predictions include all predictions that match the gold label and all predictions that clearly indicate a grammatical gender. Underspecified predictions are all predictions that form a grammatical sentence, but do not uniquely encode a grammatical gender (i.e. forms that are compatible with more than one gender), see Section 5.1. Examples of this would be the indefinite determiner *ein*, which is used for both the neuter and masculine nouns. An exception is made for plural predictions that correspond to the gold determiner; these are counted as underspecified predictions because they do not allow the grammatical gender of the loanword to be inferred. Incorrect predictions are all those, that infer an incorrect grammatical gender or do not form a grammatically correct sentence.

5.1 Alternative Predictions

For this experimental setup, the definite nominative (and in some cases the accusative) determiner is used as the gold label. However, there are alternative forms that express gender and form a grammatically correct sentence. A summary of the alternative predictions considered for each gender is provided in Table 5.1. These alternatives include possessive, demonstrative, and distributive pronouns, as well as indefinite determiners.

Gender	Nom	Acc	Ind	Possessive	Dem	Distr
Masculine	der	/	ein	mein/dein/sein/ihr/unser/euer	dieser	jeder
Feminine	die	die	eine	meine/deine/seine/ihre/unsere/eure	diese	jede
Neuter	das	das	ein	mein/dein/sein/ihr/unser/euer	dieses	jedes

Table 5.1: Alternatives by Gender and Determiner Type. Masculine accusative forms are not included in the dataset and are marked with "/".

For all instances in which the loanword occurs in the accusative case, the loanword belongs to either the feminine or the neuter gender, as their accusative determiners (*die*, *das*) agree with the nominative forms, while the masculine accusative *den* does not and is therefore excluded.

Therefore, the indefinite, possessive, demonstrative, and distributive determiners that may occur as alternatives are the same as those used for feminine and neuter loanwords in the nominative case.

5.1.1 Definition of Correct, Underspecified, and Incorrect Predictions

Correct predictions are all those corresponding to gold, or those that clearly indicate an acceptable grammatical gender. Given Table 5.1, the alternative predictions that clearly indicate one grammatical gender are all demonstrative and distributive pronouns (for all three genders), as well as all female possessive and feminine indirect determiners (when the target is the feminine). These are counted as correct predictions if they match the gender of the gold label

Masculine and neuter indefinites and possessive pronouns presented in Table 5.1 are counted as underspecified predictions, as they do not uniquely encode grammatical gender. In addition, the undespecification category includes lexical items listed in a predefined exclusion set (e.g. proper names such as *Mr* or *John*) as well as determiners that are ambiguous across multiple genders according to the mapping in Table 5.1. The predefined exclusion set was manually created based on the incorrect predictions that still formed grammatical sentences. Example 9 shows the prediction of a proper name (*Jessica*) for a masked determiner, which would be counted as underspecified.

(9) Jessica Chairwoman hat ihren Stift fallen lassen.
Jessica chairwoman have.3SG.PRS 3SG.F.POSS pen drop let
'Jessica(the) chairwoman dropped her pen.'

Other predictions, such as determiners that correspond to an incorrect gender, or predictions that do not make grammatical sentences are counted as incorrect. An example of this would be 10 for which mBERT predicted 11.

(10) [MASK] Charity war ein voller Erfolg.
[MASK] charity be.PST.3SG a complete success
DEF.F.SG.NOM charity COP.PST.3SG INDF.M.SG.NOM complete success

"The charity event was a complete success."

(11) Charity Charity war ein voller Erfolg.
charity charity be.PST.3SG a complete success
"Charity charity was a complete success."

It should also be noted that approximately 30 loanwords appear for more than one subclass because they fit multiple gender assignment patterns. No special distinction or separation was made for these duplicates, and they are treated as independent instances across the results.

5.2 Overall Results

The main results can be seen in Table 5.2. The number of correct predictions increases for each model and category after finetuning. Furthermore, none of the models make underspecified predictions after finetuning. Incorrect predictions decrease across all

models and classes after fine-tuning, with the exception of gBERT for the biological gender category, where the proportion of incorrect predictions increases slightly from 0.021 in the zero-shot setting to 0.042 after fine-tuning. Further differences across classes, subclasses and models will be described in the following sections.

In the zero-shot setting, translation was consistently the most difficult class, while biological gender and plurals achieved the highest accuracies. After fine-tuning, performance improved across all classes and model differences became substantially smaller, with several classes approaching or reaching ceiling performance. This will further be discussed in Section 5.3.

5.3 Performance by Class

This section will analyse the accuracy results in the three-way split by the five classes in the dataset: morphological mapping, biological gender, translation, phonological mapping and plurals. The plurals are included in the results for completeness, though they will be interpreted separately in the analysis, as they cannot be used as a proxy for grammatical gender.

5.3.1 Class Performance Zero-Shot

The morphological mapping class achieves correct predictions percentages ranging from 0.676 (mBERT) to 0.878 (gBERT), the highest performance across all classes in the zero-shot setting. dbmdzBERT (0.838) ranks second to the performance of gBERT, followed by GermanBERT (0.824). For this class underspecified predictions are rare across all models (0.027–0.054). Overall, morphological mapping appears to be a well-handled class in the zero-shot setting, with all monolingual German models exceeding 0.80 accuracy.

Zero-shot predictions for the biological gender subclass are consistently strong across models, with minor differences in the percentage of correct predictions, which range from 0.833 (mBERT) to 0.875 (gBERT and dbmdzBERT). Incorrect predictions are very low across all models; mBERT has the highest percentage of incorrect predictions (0.083). GermanBERT does not make any incorrect predictions but has the highest percentage of underspecified predictions (0.146).

The translation-based class is the most challenging class for models in the zero-shot setting. It is the only class where models fall below 0.500 accuracy (excluding plurals). Correct predictions range from only 0.459 (mBERT) to 0.803 (dbmdzBERT), with the largest performance gap of 0.344 accuracy points. mBERT has the highest amount of incorrect predictions for this class (0.426), while dbmdzBERT has the lowest amount (0.131). Underspecified predictions remain comparatively frequent (0.066–0.115)

For the phonological subclass correct predictions range between 0.696 (dbmdzBERT) and 0.826 (gBERT). gBERT achieves the strongest performance (highest correct scores), followed by germanBERT (0.739) and mBERT (0.717). The highest rate of underspecified predictions is observed for dbmdzBERT (0.130). Overall, phonological cues appear easier than translation but more difficult than biological gender.

Lastly, the models perform very strongly on plural nouns in the zero-shot setting. mBERT produces two incorrect predictions, while GermanBERT produces one. All remaining predictions correspond to the gold determiner. However, these results should be interpreted with caution, as the plural, the determiner *die* is used for nouns of all

three grammatical genders. Consequently, correct predictions of *die* do not indicate one grammatical gender and are therefore classified as underspecified. Despite this limitation, the models consistently generate grammatically correct plural determiners, resulting in strong overall performance for this class.

Class	**Metric**	**mBERT**		**GermanBERT**		**gBERT**		**dbmdzBERT**	
		Zero	Fine	Zero	Fine	Zero	Fine	Zero	Fine
Morphological Mapping	Correct	0.676	0.946	0.824	0.946	0.878	0.959	0.838	0.959
Morphological Mapping	Underspecified	0.054	/	0.027	/	0.027	/	0.054	/
Morphological Mapping	Incorrect	0.270	0.054	0.149	0.054	0.095	0.041	0.108	0.041
Biological Gender	Correct	0.833	1.000	0.854	1.000	0.875	0.958	0.875	0.979
Biological Gender	Underspecified	0.083	/	0.146	/	0.104	/	0.104	/
Biological Gender	Incorrect	0.083	/	/	/	0.021	0.042	0.021	0.021
Translation	Correct	0.459	0.721	0.689	0.721	0.787	0.869	0.803	0.902
Translation	Underspecified	0.115	/	0.115	/	0.066	/	0.066	/
Translation	Incorrect	0.426	0.279	0.197	0.279	0.148	0.131	0.131	0.098
Phonological	Correct	0.717	0.978	0.739	0.978	0.826	0.957	0.696	0.957
Phonological	Underspecified	0.065	/	0.065	/	0.043	/	0.130	/
Phonological	Incorrect	0.217	0.022	0.196	0.022	0.130	0.043	0.174	0.043
Plurals	Correct	/	/	/	/	/	/	/	/
Plurals	Underspecified	0.900	1.000	0.950	1.000	1.000	1.000	1.000	1.000
Plurals	Incorrect	0.100	/	0.050	/	/	/	/	/
Macro Average	Correct	0.671	0.911	0.777	0.911	0.842	0.936	0.803	0.949
Macro Average	Underspecified	0.104	/	0.089	/	0.060	/	0.089	/
Macro Average	Incorrect	0.225	0.089	0.134	0.089	0.098	0.064	0.108	0.051

Table 5.2: Accuracy Model Performance by Class and Prediction Type (Corres, Underspecified or Incorrect) Across Zero-Shot and Finetuned Settings. A slash (/) indicates that no predictions of the respective type were produced

5.3.2 Class Performance Finetuned

After finetuning, none of the models make underspecified predictions for any class.

The performance for the morphological mapping subclass improves across all models. Correct predictions increase to 0.946 (mBERT and GermanBERT) to 0.959 (gBERT and dbmdzBERT) across all models. The percentage of incorrect predictions decreases to 0.041–0.054, with a minimal performance difference across models (0.013).

Biological gender remains one of the best performing classes after finetuning. mBERT and GermanBERT achieve perfect performance for correct (1.000), dbmdzBERT reaches 0.979, while gBERT obtains 0.958. Incorrect predictions are rare (0.000–0.042). The results suggest that biological gender information can be learned almost perfectly through fine-tuning.

The translation-based class benefits from finetuning but remains the most difficult class. Correct predictions increase from 0.459–0.803 in zero-shot to 0.721–0.902 after fine-tuning. dbmdzBERT achieves the highest performance (0.902 correct), followed by gBERT (0.869 correct). Translation remains the only class where none of the models exceed 0.95 accuracy.

The phonological subclass fine-tuning produces near-ceiling performance for all models. Correct prediction rates range from 0.957 (gBERT and dbmdzBERT) to 0.978 (mBERT and GermanBERT). Incorrect predictions fall to 0.022–0.043.

For the plural class all models predict the gold determiner after finetuning. As mentioned above, the plural determiners do not serve as a proxy for grammatical gender and are therefore counted as underspecified predictions.

5.4 Performance by Subclass

The five classes in the training data can be divided into their subclasses, as introduced in Section 3.2. The current Section analyses performance by subclass, which is presented in Table 5.3.

It should be noted that underspecification counts are not directly comparable across all classes. Some classes are linked to the feminine gender, for which all alternative determiners introduced in Table 5.1 (e.g., *eine*, *meine*) unambiguously indicate feminine gender and are therefore counted as correct predictions. In contrast, masculine and neuter alternatives (e.g., *ein*, *mein*) are compatible with more than one gender and are thus classified as underspecified when they are predicted.

Because of this, classes with the feminine gold label tend to show lower underspecification rates, even if the underlying model behaviour is comparable across classes. For example, a prediction such as *eine* for a feminine noun is counted as correct because it uniquely identifies the feminine gender. In contrast, a prediction such as *ein* for a masculine or neuter noun is counted as underspecified because it can correspond to both genders. The underspecified predictions for grammatically feminine nouns mainly consist of proper name predictions, such as shown in the prediction 13 for 12. Subclasses with the feminine gender are *-ity*, *-ion*, *-ness*, as well as the feminine translation and biological gender subclasses.

(12) [MASK] Chairwoman hat ihren Stift fallen lassen.
DEF.F.SG.NOM chairwoman have.PRS.3SG her pen drop.INF let.PTCP
DEF.F.SG.NOM chairwoman AUX.3SG POSS.M.SG.ACC pen drop.INF let.PTCP

"The chairwoman dropped her pen."
Gold: *die*

(13) Jessica Chairwoman hat ihren Stift fallen lassen.
Jessica chairwoman have.PRS.3SG her pen drop.INF let.PTCP
"Jessica Chairwoman dropped her pen."
$\text{mBERT}_{\text{zero}}$: *Jessica*

5.4.1 Subclass Performance Zero-Shot

Zero-shot performance varies strongly across subclasses. These variations will be highlighted in the current section.

The morphological subcategory *-er* (masculine) has an overall strong performance in the zero-shot setting. mBERT achieves 1.000 correct predictions, while the monolingual German models show slightly lower performance (0.800–0.867 correct). Errors and underspecified predictions are minimal (see Table 5.3). The *-ing* subclass (neuter) is one of the weakest morphological subclasses in zero-shot. mBERT performs particularly low with only 0.348 correct and 0.565 incorrect. Other models perform substantially better (GermanBERT and dbmdzBERT: 0.826, gBERT: 0.913). Underspecified predictions are low across models (0.043 for all monolingual models and 0.087 for mBERT). For the *-ion* subcategory, all models achieve a generally high performance in the zero-shot setting, with accuracy ranging from 0.846 (gBERT) to ceiling performance (GermanBERT and dbmdzBERT). Underspecified predictions are very rare for this subclass, likely because it corresponds to the feminine gender. Most models achieve high zero-shot accuracy for the *-ity* subclass, ranging from 0.769 (mBERT) to 1.000 (gBERT). The error rate (incorrect predictions) are higher than that for the *-ion*, while the percentage of underspecified predictions is constant with those for the *-ion* subcategory. Lastly, the *-ness* subcategory has relatively low zero-shot accuracy compared to that for the other morphological mapping subcategories. Accuracy ranges from 0.500 (mBERT and GermanBERT) to 0.700 (gBERT). There are no underspecified predictions. Overall, this subclass shows a relatively high average proportion of incorrect predictions across models.

The biological gender subclasses both show strong performance in the zero-shot setting. The biological feminine subclass has slightly higher accuracy scores for all models. For this subclass, accuracy ranges from 0.875 (mBERT) to 0.958 (dbmdzBERT). For the masculine subclass, accuracy ranges from 0.792 (mBERT, germanBERT and dbmdzBERT) to 0.833 (gBERT). The incorrect predictions are higher for the feminine subclass (when there are any). This discrepancy is caused by the underspecified predictions, which are higher for the masculine subcategory, as the feminine subcategory is less prone to underspecification.

For the translation-based subclasses, performance varies. The masculine translation-based subclass shows the highest accuracy scores, despite being subject to more possibilities for underspecified predictions than the feminine subclass. mBERT shows the lowest zero-shot accuracy for this class (0.667), while gBERt and dbmdzBERT show the highest performance (0.810). Performance for the feminine subclass is lower than that for the masculine subclass across all models. mBERT has the lowest percentage of correct predictions (0.500), while all monolingual German models achieve a correct prediction percentage of 0.750. Underspecification is lower for the monolingual German

Subclass	**Metric**	**mBERT**		**GermanBERT**		**gBERT**		**dbmdzBERT**	
		Zero	Fine	Zero	Fine	Zero	Fine	Zero	Fine
-er (M)	Correct	1.000	1.000	0.867	1.000	0.867	1.000	0.800	1.000
	Underspecified	/	/	0.067	/	0.067	/	0.200	/
	Incorrect	/	/	0.067	/	0.067	/	/	/
-ing (N)	Correct	0.348	1.000	0.826	1.000	0.913	1.000	0.826	1.000
	Underspecified	0.087	/	0.043	/	0.043	/	0.043	/
	Incorrect	0.565	/	0.130	/	0.043	/	0.130	/
-ion (F)	Correct	0.923	0.923	1.000	0.923	0.846	1.000	1.000	1.000
	Underspecified	0.077	/	/	/	/	/	/	/
	Incorrect	/	0.077	/	0.077	0.154	/	/	/
-ity (F)	Correct	0.769	0.923	0.846	0.923	1.000	1.000	0.923	1.000
	Underspecified	0.077	/	/	/	/	/	/	/
	Incorrect	0.154	0.077	0.154	0.077	/	/	0.077	/
-ness (F)	Correct	0.500	0.800	0.500	0.800	0.700	0.700	0.600	0.700
	Underspecified	/	/	/	/	/	/	/	/
	Incorrect	0.500	0.200	0.500	0.200	0.300	0.300	0.400	0.300
Biological (F)	Correct	0.875	1.000	0.917	1.000	0.917	0.917	0.958	0.958
	Underspecified	0.042	/	0.083	/	0.042	/	/	/
	Incorrect	0.083	/	/	/	0.042	0.083	0.042	0.042
Biological (M)	Correct	0.792	1.000	0.792	1.000	0.833	1.000	0.792	1.000
	Underspecified	0.125	/	0.208	/	0.167	/	0.208	/
	Incorrect	0.083	/	/	/	/	/	/	/
Translation (F)	Correct	0.500	0.700	0.750	0.700	0.750	0.850	0.750	0.850
	Underspecified	0.100	/	0.050	/	0.050	/	/	/
	Incorrect	0.400	0.300	0.200	0.300	0.200	0.150	0.250	0.150
Translation (M)	Correct	0.667	0.905	0.762	0.905	0.810	1.000	0.810	1.000
	Underspecified	0.095	/	0.143	/	0.095	/	0.143	/
	Incorrect	0.238	0.095	0.095	0.095	0.095	/	0.048	/
Translation (N)	Correct	0.200	0.550	0.550	0.550	0.800	0.750	0.850	0.850
	Underspecified	0.150	/	0.150	/	0.050	/	0.050	/
	Incorrect	0.650	0.450	0.300	0.450	0.150	0.250	0.100	0.150
Final Plosive (M)	Correct	0.696	1.000	0.652	1.000	0.739	1.000	0.609	0.957
	Underspecified	0.087	/	0.087	/	0.087	/	0.174	/
	Incorrect	0.217	/	0.261	/	0.174	/	0.217	0.043
Monosyllabic (M)	Correct	0.739	0.957	0.826	0.957	0.913	0.913	0.783	0.957
	Underspecified	0.043	/	0.043	/	/	/	0.087	/
	Incorrect	0.217	0.043	0.130	0.043	0.087	0.087	0.130	0.043

Table 5.3: Subclass Performance Across Models in Zero-Shot and Fine-Tuned Settings.
A slash (/) indicates that no predictions of the respective type were produced

models, while it is higher in mBERT, despite there being fewer underspecifying alternatives for this feminine subclass. Performance for the neuter subclass varies strongly across models, with an accuracy of 0.200 for mBERT and 0.850 for dbmdzBERT. All models have some underspecified predictions ranging from 0.050 (gBERT and dbmdzBERT) to 0.150 (mBERT and germanBERT).

The phonological subclasses show slight variation. Performance for the final plosive subcategory is slightly higher when looking at correct predictions, while it is lower when looking at incorrect predictions. This means that there were more underspecified predictions for the final plosive subclass than for the monosyllabic subclass in the zero-shot setting. For the final plosive subclass, accuracy ranges from 0.609 (dbmdzBERT) to 0.739 (gBERT). dbmbdzBERT also shows the highest percentage of underspecified predictions for this subclass. For the monosyllabic subclass zero-shot predictions are relatively strong, with accuracy ranging from 0.739 (mBERT) to 0.913 (gBERT). The amount of underspecified predictions is lower than for the final plosive subcategory, while the number of incorrect predictions is the same (mBERT) or lower.

The plural class does not have subclasses. Therefore, this section will not further report on predictions made for it.

5.4.2 Subclass Performance Finetuned

Finetuning leads to near-ceiling performance for many subclasses. Underspecified predictions are eliminated for all models and all classes after finetuning. This section further reports the performance results of the four models on the different subcategories after finetuning.

For the morphological mapping subcategory *-er* all models achieve ceiling performance after finetuning, indicating this subcategory is highly learnable through finetuning. The same goes for the *-ing* subcategory. For the *-ion* subcategory, all models except mBERT achieve ceiling performance. The percentage of correct predictions for mBERT does not improve compared to its zero-shot performance. The *-ity* subcategory shows strong overall performance, with mBERT (0.923 correct) and germanBERT (0.928 correct) on the weaker end, while gBERT and dbmdzBERT achieve ceiling performance. Lastly, the *-ness* subcategory is the weakest of this class across models. mBERT and germanBERT improve their correct predictions by 0.300 points to 0.800. gBERTs performance stays constant, while dbmbdzBERT improves its correct predictions by 0.100 points to 0.700. These results indicate that the morphological mapping subcategories are highly learnable during finetuning.

mBERT and germanBERT achieve ceiling performance for both the biological feminine and the biological masculine subclasses after finetuning. gBERT and dbmdzBERT only achieve ceiling performance for the masculine biological gender subcategory. For the feminine subcategory, they achieve near ceiling performance (gBERT: 0.917, dbmdzBERT: 0.958). There is a strong convergence towards correct predictions after finetuning for both subclasses. translation-basedtranslation-basedtranslation-based For the translation based subclasses, the results are varied across subclasses and models. The masculine translation based subclass is among the strongest subclasses after finetuning, with the percentage of correct predictions ranging from 0.905 (mBERT and germanBERT) to ceiling performance (gBERT and dbmdzBERT). The feminine subclass shows moderate improvement, ranging from 0.700 correct (mBERT and germanBERT) to 0.850 correct (gBERT and dbmdzBERT). The neuter subclass remains the weakest across the three subclasses. mBERT shows an increase from 0.200 correct (zero-shot)

to 0.500 correct (finetuned). Interestingly, germanBERT and dbmdzBERT do not show higher percentages of correct predictions than they did in the zero-shot setting, while their percentages of incorrect predictions increase. The accuracy for gBERT decreases after finetuning, while the percentage of incorrect predictions increases.

Both phonological subclasses show strong improvements after finetuning. For the final plosive subclass, all models except dbmdzBERT reach ceiling performance. dbmdzBERT improves its performance from 0.609 to 0.957. The results for the monosyllabic subclass are slightly lower. mBERT, germanBERT and dbmdzBERT have a percentage of correct results of 0.957, while gBERT's performance does not improve and stay at an accuracy of 0.913.

As mentioned in the previous subcategory, the plurals class does not have subclasses and is therefore not further discussed in this section.

5.5 Gender Based Performance

Each subclass corresponds to a single grammatical gender; the performance scores can be macro-averaged per grammatical gender for each model. These scores are reported in Table 5.4. The plurals subclass is excluded, as it does not allow for gender distinction based on the determiner.

Gender	**Metric**	**mBERT**		**GermanBERT**		**gBERT**		**dbmdzBERT**	
		Zero	Fine	Zero	Fine	Zero	Fine	Zero	Fine
Feminine	Correct	0.725	0.875	0.825	0.875	0.850	0.900	0.863	0.912
	Underspecified	0.062	/	0.037	/	0.025	/	/	/
	Incorrect	0.212	0.125	0.138	0.125	0.125	0.100	0.138	0.087
Masculine	Correct	0.764	0.972	0.774	0.972	0.830	0.981	0.755	0.981
	Underspecified	0.075	/	0.113	/	0.094	/	0.160	/
	Incorrect	0.160	0.028	0.113	0.028	0.075	0.019	0.085	0.019
Neuter	Correct	0.279	0.791	0.698	0.791	0.860	0.884	0.837	0.930
	Underspecified	0.116	/	0.093	/	0.047	/	0.047	/
	Incorrect	0.605	0.209	0.209	0.209	0.093	0.116	0.116	0.070

Table 5.4: Macro-averaged Performance by Grammatical Gender Across Subclasses. A slash (/) indicates that no predictions of the respective type were produced

The results show that fine-tuning improves correct predictions across all models and genders, except for neuter predictions by gBERT. Underspecified predictions vanish completely after finetuning. Predictions for the masculine gender show the highest percentage of correct predictions after finetuning, for all models. Predictions for the feminine gender are already strong in the zero-shot setting. Predictions for the neuter gender have the lowest percentage of correct predictions in the zero-shot setting, especially for mBERT(0.279). The remaining errors after finetuning are mostly related to loanwords associated with the neuter gender.

5.6 Model Comparison

This section compares the results by models. In the zero-shot setting, mBERT shows the largest variability across subclasses. Its performance is perfect for the *-er* subclass,

and relatively high for the biological gender subclasses (0.875 correct for feminine and 0.792 correct for masculine). However, it struggles strongly with several other categories. In particular, the translation subclasses show the model's weaknesses, with 0.500 correct for the feminine translation subcategory and only 0.200 correct for the neuter translation subcategory. The *-ing* subclass is also difficult, with only 0.348 correct predictions and a high percentage of incorrect predictions (0.565). After fine-tuning, mBERT improves substantially across nearly all subclasses, reaching 1.000 correct predictions for *-er* and *-ing*. However, weaker performance remains for the *-ness* subclass (0.800) and neuter translation subclass (0.550), indicating that some lexical or semantic ambiguity persists even after fine-tuning.

GermanBERT shows a more balanced zero-shot performance across subclasses. It achieves high performance for several categories, including *-ion* (1.000 correct) and the biological gender subclasses (0.917 correct for feminine and 0.792 correct for masculine). Performance is lower for some morphological subclasses, in particular *-ness* (0.500) correct, as well as the translation-based subclasses, where the percentage of correct predictions ranges from 0.550 (neuter translation) to 0.750 (feminine translation). Incorrect predictions remain low overall, but are still present in subclasses such as *-ing* (0.130 incorrect) and *-ity* (0.154 incorrect), while underspecified predictions occur mainly for the biological gender subclasses (0.083 for feminine and 0.208 for masculine). After fine-tuning, GermanBERT reaches ceiling performance for several subclasses, including *-er*, *-ing*, and both biological gender subclasses. Performance remains slightly lower for *-ness* (0.800 correct) and the neuter translation subclass (0.550 correct), while most other subclasses reach between 0.923 and 1.000 accuracy.

gBERT shows strong and relatively stable zero-shot performance across most subclasses. It achieves perfect results for *-ity*, and very high performance for *-ing* (0.913 correct), and both biological gender subclasses (0.917 feminine, 0.833 masculine). Lower performance is observed for *-ness* (0.700 correct) and the translation-based subclasses, where the percentage of correct predictions ranges from 0.750 (feminine) to 0.810 (masculine and neuter). Underspecified predictions are generally rare (0.000–0.067), while incorrect predictions are mainly concentrated in *-ing* and translation-based subclasses. After fine-tuning, gBERT reaches 1.000 correct for many subclasses, such as *-er* or translation (M). The subclass *-ness* only reaches 0.700 correct, the biological feminine gender (0.917), the monosyllabic subcategory (0.913) and the feminine (0.850) and neuter (0.750) translation subclass remaining below ceiling.

dbmdzBERT shows generally strong zero-shot performance across subclasses. It achieves perfect performance for *-ion* (1.000 correct) and good scores for the biological gender subclasses (0.958 feminine, 0.792 masculine). Lower performance is observed for *-ness* (0.600 correct) and *-er* (0.800 correct). The *-ing* subclass performs relatively well (0.826 correct), though there are some incorrect predictions (0.130 incorrect). The translation-based subclasses also show variation, ranging from 0.850 (neuter) to 0.750 (feminine). Underspecified predictions are rare overall but appear mainly in the biological masculine subclass (0.208 underspecified) and the *-er* subclass (0.200 underspecified). After fine-tuning, dbmdzBERT reaches ceiling performance for almost all morphological mapping subclasses, the masculine translation-based subclass and the masculine biological gender subclass. The performance for the *-ness* (0.700 correct) subclass and the translation-based subclasses (0.850 across genders) remain lower, even after finetuning.

Overall, gBERT and dbmdzBERT show the strongest zero-shot performance across

subclasses, with both models reaching high or near-ceiling results for several subcategories, in particular *-ity*, *-ing*, and *-ion*, and the biological gender subclasses. GermanBERT performs at a similar level overall but shows more variation across subclasses, with lower results in categories such as the translation-based subclasses. mBERT shows the lowest overall performance, with particularly weak results for *-ing* (0.348 correct) and the translation subclasses, in the zero-shot setting, while still performing strongly in more regular patterns such as *-er* (1.000 correct) and the biological gender subclasses after finetuning.

5.7 SEED Results

A multiple-seed experiment was done to ensure that model performance is not influenced by randomness. Each model was trained using four different random seeds. The performance of each model across different random seeds, evaluated using the three-way prediction split introduced above, is presented in Table 5.5. For simplicity, only the proportion of correct predictions is reported. The underspecified and incorrect proportions are reported in the Appendix A.6.1.

Seed	mBERT	GermanBERT	gBERT	dbmdzBERT
33	0.912	0.912	0.940	0.952
42	0.900	0.900	0.936	0.936
123	0.892	0.876	0.944	0.932
3210	0.888	0.892	0.948	0.932
Mean	0.898	0.894	0.942	0.937
Standard Deviation	±0.009	±0.012	±0.004	±0.008

Table 5.5: Correct Prediction Rates Across Different Random Seeds

Overall, performance remains stable across seeds for all models. gBERT shows the lowest standard deviation (±0.004), followed by dbmdzBERT (±0.008) and mBERT (±0.009). GermanBERT shows the largest variation across seeds (±0.012), though the differences remain small overall. These results suggest that the reported performances are not strongly dependent on random initialisation.

5.8 Confidence Scores

For each masked token, the model outputs logits over the vocabulary. These logits are converted into probabilities using the softmax function, creating confidence scores for each prediction. The confidence scores for correct, underspecified and incorrect predictions are presented in Table 5.6.

	mBERT		**GermanBERT**		**gBERT**		**dbmdzBERT**	
	Zero	Fine	Zero	Fine	Zero	Fine	Zero	Fine
Correct	0.593	0.994	0.679	0.994	0.641	0.998	0.667	0.994
Underspecified	0.267	/	0.458	/	0.515	/	0.514	/
Incorrect	0.391	0.964	0.497	0.964	0.470	0.935	0.515	0.978

Table 5.6: Model Performance by Prediction Type in Zero-Shot and Fine-Tuned settings

The confidence scores indicate that correct predictions have higher confidence scores than underspecified predictions for all models. In the zero-shot setting, the confidence scores for incorrect predictions are slightly higher than those for underspecified predictions for most models. mBERT has the lowest confidence across all three prediction types in the zero-shot setting (0.267-0.593), whereas the monolingual German models show higher confidence scores, ranging from 0.458 (GermanBERT underspecified) to 0.679 (germanBERT correct). Finetuning increases the confidence for both correct and incorrect predictions, where correct predictions always have slightly higher accuracy. The confidence scores will further be investigated in the analysis Chapter 6.

Chapter 6

Analysis

This Chapter provides a detailed analysis of the predictions made by the four models, both before and after finetuning. The focus lies on the models abilities to correctly predict gender, taking alternative predictions into account. Possible alternative predictions have been introduced in Section 5.1.

In contrast to the Results chapter, predictions are not divided into correct, underspecified, and incorrect categories. Instead, all alternative predictions presented in Table 5.1 are considered valid if they indicate the same gender as the gold label. As a result, there is no separate underspecification category in the analysis. Predictions of proper names are counted as incorrect, as they do not provide insights in the model's preference for a grammatical gender.

This Chapter first evaluates model accuracy under this broader definition of valid predictions, followed by an error analysis of all invalid predictions. Finally, other potential influencing factors are discussed.

6.0.1 Prediction of Acceptable Alternatives

The overall number of valid alternatives per model and setting (zero-shot or finetuned) is summarised in Table 6.1. As reported in the results section, the total number of errors decreases after fine-tuning.

	mBERT		germanBERT		gBERT		dbmz	
	Zero	Fine	Zero	Fine	Zero	Fine	Zero	Fine
Total errors	98	22	60	22	44	15	50	12
Acceptable alternatives	9	0	23	0	19	0	23	0
Acceptable alternatives (%)	9.2%	0.0%	38.3%	0.0%	43.2%	0.0%	46.0%	0.0%

Table 6.1: Predictions of Acceptable Alternatives

In the zero-shot setting, a notable proportion of the non-gold predictions (in comparison to gold) correspond to accepted alternative predictions 6.1. This is particularly true for the monolingual German models. Additionally, it should be noted that mBERT predicted *"the"* as a determiner 12 times in the zero-shot setting. These cases are excluded from the valid alternatives as the English determiner *"the"* does not encode grammatical gender. After finetuning, none of the models made alternative predictions. The effect of finetuning will be discussed in the following section.

6.0.2 Effects of Finetuning on Alternative Predictions

After finetuning, the acceptable alternatives disappear completely. This suggests that finetuning trains the models to predict the nominative (and in some cases the accusative) definite determiner, the only types of determiners seen during fine-tuning. After finetuning, almost all valid alternatives predicted in the zero-shot setting changed to the gold determiner. Table 6.2 shows how many valid alternative predictions a model made in the zero-shot setting became the gold prediction after fine-tuning.

No invalid prediction in the zero-shot setting becomes a valid alternative after fine-tuning. For mBERT and dbmdzBERT, all loanwords for which a valid alternative was predicted in the zero-shot setting are assigned the gold determiner after fine-tuning. gBERT and germanBERT switch from valid alternative predictions to incorrect direct determiners for several predictions.

Model	**Number of Alternatives to Gold**
mBERT	9/9
germanBERT	21/23
gBERT	18/19
dbmdzBERT	23/23

Table 6.2: Number of Correct Alternative Predictions per Model

There are three instances where models go from predicting the correct gender (through an alternative prediction) in the zero-shot setting to an incorrect gender prediction after finetuning. GermanBERT switches from valid alternatives for *Firewall* (feminine) and *Abortion* (feminine), to invalid direct nominative (or accusative) determiners (*der*) (masculine) and *das* (neuter), respectively. gBERT does the same for *Dope* (neuter), where it switches from the valid alternative *ein* (indefinite neuter) to *die* (feminine). Neither the surrounding sentences nor the loanwords themselves provide a clear indication of why these two errors occur. Overall, these results suggest that the models learn to prioritise the prediction of the gold definite determiners during fine-tuning, rather than capturing more general grammatical gender patterns.

6.0.3 Confidence for Alternative Predictions

The confidence scores are reported for the highest-ranked prediction only. The predictions are grouped into three categories: gold predictions, valid alternative predictions, and invalid predictions. The confidence scores for these prediction types are reported in Table 6.3. The values in the table indicate the average confidence score, while the numbers in brackets represent the number of instances. After fine-tuning, no valid alternative predictions remain. The table shows an increase in confidence for invalid predictions after fine-tuning.

	mBERT		germanBERT		gBERT		dbmz	
	Zero	Fine	Zero	Fine	Zero	Fine	Zero	Fine
Gold	0.61 (146)	0.99 (207)	0.70 (169)	0.99 (207)	0.65 (185)	0.99 (214)	0.68 (179)	0.99 (217)
Valid	0.36 (9)	0.00 (0)	0.56 (23)	0.00 (0)	0.62 (19)	0.00 (0)	0.58 (23)	0.00 (0)
Invalid	0.36 (74)	0.96 (22)	0.42 (37)	0.96 (22)	0.45 (25)	0.94 (15)	0.49 (27)	0.98 (12)

Table 6.3: Average Confidence of the Highest-Ranked Prediction Across Prediction Types and Settings. Values in Brackets Indicate the Number of Instances.

In the zero-shot setting, confidence scores for valid alternatives vary strongly across models. The lowest average confidence for valid alternatives is observed for mBERT (0.36), while the highest is found for gBERT (0.62). It is also noteworthy that mBERT assigns the same average confidence to valid and invalid (incorrect) predictions in the zero-shot setting. For invalid predictions in the zero-shot setting, the variation across models is not that strong. The confidence for alternative predictions in the zero-shot setting is lower than that for gold predictions.

After fine-tuning, the confidence for invalid predictions increases across all models. dbmdzBERT shows the highest confidence for invalid predictions (0.98) while gBERT has the lowest confidence (0.94). This suggests that fine-tuning leads to a stronger preference for the gold direct determiner, while alternative predictions only occur rarely.

6.0.4 Gender in Alternative Predictions

This section analyses how often the models predict any valid alternative determiner (all determiners in Table 5.1), and in how many of these cases the predicted alternative also corresponds to the correct gender. It measures how frequently a valid alternative is produced together with a correct gender assignment. As no valid alternative predictions are made after fine-tuning, the analysis in this section focuses exclusively on the zero-shot setting. Loanwords that allow for multiple genders for which an alternative prediction was made are considered correct if the alternative prediction matches one of the genders.

An incorrect alternative prediction are all cases where an alternative determiner was predicted, but for the wrong gender. In the zero-shot setting, the number of incorrect alternative predictions is very low across all models. There is a total of three cases in which this happens: once for mBERT, germanBERT and gBERT.

A closer look at these cases shows that mBERT assigns a feminine gender to the loanword “Dope” by predicting the indefinite determiner “eine”. After fine-tuning, it predicts the gold determiner *das* (neuter), which is correct. gBERT assigns the neuter indefinite determiner *ein* for the loanword *Frisbee* (feminine) and germanBERT the indefinite masculine determiner for *Stuntwoman* (feminine). The observed behaviour is likely driven by information acquired during pretraining rather than by contextual evidence in the input.

6.1 Gender Determination Accuracy

This section will report the models accuracies in reporting the correct gender for an English Loanword embedded into a German sentence. A correct prediction is one that indicates the same grammatical gender as the gold label and forms a grammatically correct sentence, under the broader evaluation setup used in this chapter. All instances from the plural class are excluded from this part of the analysis as they do not provide a proxy for grammatical gender. As established in the literature review, plural nouns in German take the determiner *die* regardless of their singular grammatical gender. A correct prediction of *die* for a plural loanword therefore reflects sensitivity to number rather than gender assignment. The overall accuracy by model for each subclass is reported in Table 6.4.

Class	Subclass	mBERT		germanBERT		gBERT		dbmdzBERT	
		Zero	Fine	Zero	Fine	Zero	Fine	Zero	Fine
Morphological Mapping	-er	1.00	1.00	0.93	1.00	0.93	1.00	1.00	1.00
Morphological Mapping	-ing	0.35	1.00	0.87	1.00	0.96	1.00	0.87	1.00
Morphological Mapping	-ion	0.85	0.92	1.00	0.92	0.85	1.00	1.00	1.00
Morphological Mapping	-ity	0.77	0.92	0.85	0.92	1.00	1.00	0.92	1.00
Morphological Mapping	-ness	0.50	0.80	0.50	0.80	0.70	0.70	0.60	0.70
Phonological	Final Plosive	0.74	1.00	0.74	1.00	0.83	1.00	0.78	0.96
Phonological	Monosyllabic	0.74	0.96	0.87	0.96	0.91	0.91	0.87	0.96
Gender	Feminine	0.88	1.00	0.92	1.00	0.92	0.92	0.96	0.96
Gender	Masculine	0.83	1.00	1.00	1.00	1.00	1.00	0.96	1.00
Translation	Feminine	0.50	0.70	0.75	0.70	0.75	0.85	0.75	0.85
Translation	Masculine	0.67	0.90	0.81	0.90	0.90	1.00	0.90	1.00
Translation	Neuter	0.35	0.55	0.70	0.55	0.85	0.75	0.90	0.85

Table 6.4: Model Performance Across Subclasses and Categories

6.1.1 Subclass Level Accuracy

Taking alternative predictions into account, gender assignment based on classes and subclasses was already relatively good and stable in the zero-shot setting for most classes and models, with relatively high performance for most morphological categories or other categories that allow for potential suffix-based mappings. For example, in the biological gender category, most feminine loanwords take the German feminine suffix marker *-in*, while most masculine loanwords receive the suffix *-er*.

After fine-tuning, the performance increases across almost all subclasses, with most morphological and phonological categories reaching near-ceiling or perfect scores across models. This indicates that these patterns are highly learnable once a supervised signal is given. In particular, for morphological categories such as suffix-based mappings (*-er*, *-ing*, *-ion*, *-ity*), the models were able to assign the correct gender (often increasing from already high zero-shot values such as 0.85–1.00 up to 0.92–1.00). This suggests that certain token combinations are highly exploitable cues for the models, which is also indicated by the high performance for biological gender categories (e.g., feminine: 0.88–0.96 in zero-shot increasing to 0.92–1.00 after fine-tuning; masculine: 0.83–1.00 already in zero-shot and reaching 1.00 after fine-tuning) and the phonological subcategory final plosive (0.74–0.83 in zero-shot, to 0.96–1.00 after fine-tuning across models).

In contrast, the translation-based categories remain the most variable even after fine-tuning, with persisting differences across models and slightly lower performance in some cases, especially for feminine and neuter translations. For example, feminine translations range from 0.50–0.75 in the zero-shot setting and increase only slightly to 0.70–0.85 after fine-tuning, while neuter translations range from 0.35–0.90 and show weaker improvements to 0.55–0.85 depending on the model.

Overall, the results show that fine-tuning largely eliminates performance gaps across models for morphological and phonological categories, while translation-based categories are difficult. Morphological patterns seem to provide the most reliable signal for gender assignment, but their advantage becomes less after supervised training.

6.1.2 Gender Level Accuracy

Table 6.5 summarises model performance across the three gender classes (masculine, feminine, neuter), without taking class or subclass distribution into account. The values

represent how often each gender is correctly predicted across the full test set. Plurals are excluded from this table, as the possible determiners do not serve as a proxy for a gender.

	mBERT		germanBERT		gBERT		dbmdzBERT	
	Zero	Fine	Zero	Fine	Zero	Fine	Zero	Fine
Masculine	0.78	0.97	0.87	0.97	0.92	0.98	0.90	0.98
Feminine	0.72	0.88	0.83	0.88	0.85	0.90	0.86	0.91
Neuter	0.33	0.79	0.79	0.79	0.90	0.88	0.88	0.93

Table 6.5: Model Performance by Grammatical Gender

Across models and settings, loanwords with the masculine gender are the most reliably predicted. Feminine loanwords show slightly lower but still consistently high accuracy. The neuter gender is the most challenging class, particularly for mBERT in the zero-shot setting.

After finetuning, gender assignment improves for all genders across all models, with one exception: gBERT's performance for the grammatical gender classification of neuter nouns decreases slightly after finetuning. This indicates that finetuning does not uniformly benefit all models equally.

The stronger performance for masculine and feminine nouns can partially be explained by the dataset and category structure. The dataset contains fewer examples of neuter instances than of feminine and masculine instances. Furthermore, neuter nouns are underrepresented in both categories that contain high generalisation patterns through suffixes (morphological mapping and biological gender).

6.1.3 Gender Confidence

The confidence scores for correctly assigned gender predictions are presented in Table 6.6. The results indicate that, in the zero-shot setting, mBERT shows significantly higher confidence for correctly assigned masculine and feminine nouns than for neuter nouns. GermanBERT, on the other hand, shows slightly higher confidence for neuter loanwords in the zero-shot setting. After fine-tuning, all models have high confidence across all genders, with little to no variation between genders.

	mBERT		germanBERT		gBERT		dbmdzBERT	
Gender	Zero	Fine	Zero	Fine	Zero	Fine	Zero	Fine
Masculine	0.605	0.996	0.667	0.996	0.649	0.997	0.664	0.995
Feminine	0.621	0.991	0.711	0.991	0.651	0.999	0.662	0.989
Neuter	0.426	0.989	0.670	0.989	0.641	0.999	0.685	0.997

Table 6.6: Model Confidence Scores by Grammatical Gender (Zero-Shot vs Finetuned)

6.1.4 Overall Model Accuracy

The overall accuracy of each model, taking alternative predictions into account, is summarised in Table 6.7. Most models achieve relatively high accuracy even in the zero-shot setting, with consistent improvements after fine-tuning.

Fine-tuning leads to only modest improvements for most models, with the largest gain observed for mBERT. German-only models mostly outperform the multilingual

	mBERT		germanBERT		gBERT		dbmdzBERT	
	Zero	Fine	Zero	Fine	Zero	Fine	Zero	Fine
Accuracy	0.677	0.904	0.838	0.904	0.891	0.934	0.882	0.948
Correct/N	155/229	207/229	192/229	207/229	204/229	214/229	202/229	217/229

Table 6.7: Overall Accuracy and Number of Correct Predictions per Model

mBERT in both settings, suggesting that language-specific pretraining is beneficial for generalisation to loanwords. Among the German models, dbmdzBERT achieves the highest and most stable performance. GBERT performs competitively, while GermanBERT only performs as well as mBERT.

6.2 Errors

Every prediction that does not match the gold label or is not one of the accepted alternatives (Table 5.1) given the gold gender is counted as an invalid prediction. This includes predictions of proper names (such as first names), predictions of determiners in different cases, determiners in other languages, and punctuation and some other German words. Across all models, there are 163 invalid predictions linked to an incorrect gender in the zero-shot setting and 71 after fine-tuning. The number of invalid predictions decreases for all models after fine-tuning. This section discusses the main error patterns observed across models and settings.

6.2.1 Errors by Class

This part of the error analysis looks at the errors made by the class, not counting valid alternatives as errors. It focuses on error patterns across classes and highlights specific examples.

For the morphological category, there is a frequent fallback to the masculine determiner *der*, especially in the zero-shot setting and particularly for loanwords in the *-ing* subcategory, which should be neuter.

For the phonological subclass, most errors are predicting the neuter gender instead of the masculine gender. Apart from that, several hallucinations occur, such as *Lu* as the determiner of *Lunch* or *Pop* as the determiner for *Pop* (referring to the music genre). One of the incorrect phonological instances that could have fallen under the biological gender category was the loanword *Expat* (masculine), which was misclassified by GermanBERT and dbmdzBERT in the zero-shot setting. This loanword does not encode biological gender, as it lacks both feminine and masculine marking, and the surrounding sentence provides no contextual gender cues. Four other instances could have fallen under the translation category; however, their determiner should have still been masculine, still making them errors.

In the biological gender subclass, errors are more structured but still show a clear tendency to default to the masculine gender. Feminine-marked forms such as *Surferin* and *Cutterin* are incorrectly labelled as *der*(masculine), showing that explicit morphological gender marking is sometimes ignored. At the same time, certain items trigger lexical substitutions or unrelated outputs, such as *the* for *Queen*, or *Sam* for *Hater*. This suggests partial recognition of named roles but unstable application of gender agreement rules. It is not the case that incorrect genders are assigned more frequently to nouns referring to female individuals that do not have the *-in* suffix.

	mBERT		germanBERT		gBERT		dbmdzBERT	
Class	Zero	Fine	Zero	Fine	Zero	Fine	Zero	Fine
Biological Gender	0.15	0.00	0.04	0.00	0.04	0.04	0.04	0.02
Phonological	0.26	0.02	0.20	0.02	0.13	0.04	0.17	0.04
Translation	0.49	0.28	0.25	0.28	0.16	0.13	0.15	0.10
Morphological Mapping	0.34	0.05	0.15	0.05	0.09	0.04	0.11	0.04

Table 6.8: Error Rates by Class (Zero-Shot vs Fine-Tuned)

6.2.2 Incorrect Predictions Across Gender

A large share of invalid predictions consists of cases where a determiner is predicted that corresponds to a different gender than the gold label. This accounts for 163 invalid zero-shot predictions and 71 fine-tuned predictions across all models. This distribution can be seen in Table 6.9.

	mBERT		germanBERT		gBERT		dbmz		Total	
	Zero	Fine	Zero	Fine	Zero	Fine	Zero	Fine	Zero	Fine
Masculine	23	3	14	3	9	2	11	2	57	10
Feminine	23	10	14	10	12	8	11	7	60	35
Neutral	28	9	9	9	4	5	5	3	46	26

Table 6.9: Incorrect predictions across grammatical gender (zero-shot vs finetuned)

The most absolutely incorrect predictions occur for the feminine gender across models. When taking the number of instances of each gender into account, the neuter gender gets classified incorrectly the most. This is likely due to its lesser representation in the training data. After finetuning the errors for most genders decrease. The only combinations for which the number of errors does not reduce are germanBERT and gBERT for the neuter gender.

There are a few cases where a model made a prediction linked to the correct grammatical gender in the zero-shot setting, and changed to an incorrect gender after finetuning. This happened for germanBERT (10 instances), mBERT (3 instances) and gBERT (1 instance). A manual check of these predictions shows that there is no underlying pattern as to why this may have happened.

6.2.3 Overgeneralisation towards Masculine Gender

A frequent error pattern is the overgeneralisation towards the masculine gender (see Table 6.10). This occurs particularly for nouns with gold gender neuter (*das*), but also for feminine nouns (*die*). In the zero-shot setting, neuter nouns are predicted as masculine 25 times across all models, with 18 of these predictions by mBERT. Feminine nouns are predicted as masculine twice for each monolingual model and four times by mBERT, in the zero-shot setting.

Especially for the phonological patterns, it is hard to identify whether the models captured the patterns during finetuning, or if they assign the masculine gender as a fallback, as this category only has masculine subcategories.

	mBERT		germanBERT		gBERT		dbmz	
	Zero	Fine	Zero	Fine	Zero	Fine	Zero	Fine
Feminine (die)	4	3	2	3	2	2	2	1
Neuter (das)	18	8	2	8	3	3	2	2

Table 6.10: Predictions of Masculine Gender for Feminine and Neuter Nouns Across Models and Settings

6.2.4 Error Types Beyond Gender Prediction

Apart from the confusion between genders, there are other incorrect predictions the models make. These include determiners from other languages, punctuation, full noun phrases, proper names, and other frequently used German words.

Foreign Language Determiners

There are several instances in which the determiner is predicted in a foreign language. Most of these errors originate from mBERT in the zero-shot setting. In particular, the model predicts *"the"* 12 times, accounting for almost half of its non-gender-related errors. This behaviour disappears after fine-tuning.

Additionally, mBERT predicts the French determiner *"le"* once for the loanword *Pellet* (neuter), which is used as an English loanword in German, but borrowed from French in English (Duden, 2018). Since the task is defined for German determiners only, such predictions are not considered valid.

GermanBERT also produces one foreign determiner prediction for the loanword *Queue* (feminine) in the zero-shot setting. In this case, the model predicts *"la"*, which would correspond to feminine gender and is also correct in French, but does not align with the German task definition.

Proper Name Predictions

In a small number of cases, the models predicted proper names or other semantically related words instead of determiners or valid alternative gender assignments. Examples of this would be *John* being predicted as the determiner for *Snow* by germanBERT. Furthermore, for the Loanword "Doorman" in the zero-shot setting, the models predicted either *Herr* (English: "Mr.") or *David* as a determiner. For the loanword *Chairwoman*, germanBERT predicted *Frau* and gBERT *Jessica* in the zero-shot setting. There is also a hallucination, where mBERT predicted *Henry* as a determiner for *Corner*. All of these alternative predictions were made by mBERT in the zero-shot setting.

Incorrect Use of Alternatives

For all previous analyses, an alternative prediction was considered valid if it corresponded to the gold gender. Some predictions use determiners listed as valid alternatives (Table 5.1) for the wrong gender. These cases are not always separable as the indefinite determiner *ein* is used for both masculine and neuter nouns, which does not provide insights into whether a model encodes the noun as masculine or neuter (i.e. it is underspecified). Predictions of valid alternatives corresponding to the incorrect gender occur four times in total. Two cases correspond to valid alternative genders

not reflected in the gold label and are discussed further in the next section. The two remaining incorrect cases are:

- mBERT: feminine indefinite determiner predicted for *Dope* (neuter)
- germanBERT: masculine indefinite determiner predicted for *Stuntwoman*

Both occur in the zero-shot setting.

Punctuation and Dataset Inconsistencies

Three instances involve punctuation being predicted instead of determiners. mBERT predicts a full stop once (zero-shot setting), and in one shared case, all three models (mBERT, gBERT, dbmdzBERT) predict a comma (Example 14). The comma was missing in the dataset, and the sentence formed by the predictions is still grammatical without the determiner.

(14) Wir haben beschlossen [MASK] Insourcing einzustellen.
we have decided [MASK] insourcing stop.INF
1PL AUX.PRS.PRF decide.PTCP that/COMP insourcing INF-stop

"We have decided to stop insourcing."

Lexical Ambiguity

There were some examples of lexical ambiguity across the models. In some cases, a determiner of a different case was used, it did not correspond to the correct gender. Some predictions used determiners with an incorrect case, resulting in ungrammatical sentences. The four instances in which this occurred not only use the wrong case but also refer to a different gender than the gold.

One of these four cases is a loanword used in the accusative case. The loanword *break* takes both the neuter and the masculine determiner, depending on its interpretation. In the dataset, it is part of the translation-based category; however, the loanword can have two meanings: a) to take a break (break.1) and b) to break through (break.2). Sense *a* (break.1) would take the feminine gender based on the translation *die Pause*, while sense *b* (break.2) would take the masculine gender, based on the translation *der Durchbruch*. In the context in which *Break* occurred in the dataset, both interpretations are possible, and the prediction of the accusative masculine determiner, therefore, is not essentially wrong.

(15) Wir haben uns die Break wirklich verdient.
we have.1PL REFL ACC.DEF break.1 really deserve.PTCP
1PL AUX.PRS.PRF REFL ACC.DEF break.1 really deserve.PTCP
"We really deserved the break."

Figure 6.1: Sense A

(16) Wir haben uns den Break wirklich verdient.
we have.1PL REFL ACC.M.DEF break.2 really deserve.PTCP
1PL AUX.PRS.PRF REFL ACC.M.DEF break.2 really deserve.PTCP
"We really deserved the break."

Figure 6.2: Sense B

Biological Gender and Gender Marking

Some errors occur in predicting the grammatical gender for biological female references. While models generally perform well in this category, errors remain for certain cases. The models learn the suffix-based feminine marker *-in* after fine-tuning. For example, *Cutterin* and *Surferin* are correctly predicted as feminine after fine-tuning, while mBERT initially predicts masculine forms.

All German-trained models correctly generalise to feminine forms in most cases, likely due to stronger morphological knowledge from pretraining. However, *Chairwoman* remains problematic: germanBERT and gBERT predict female references (*Mrs* and *Jessica*) in zero-shot, while gBERT switches to masculine after fine-tuning, germanBERT predicts the gold feminine nominative determiner after fine-tuning. For most biologically female loanwords without explicit suffix marking, gender is correctly predicted, with the exception of *Queen* in zero-shot, for which mBERT predicts the English determiner *the* and *Chairwoman*.

Contextually, three out of 24 instances in this category allow gender inference through surrounding information (e.g. pronouns or named entities such as Nadine Angerer). However, models still fail in some of these cases, including *Chairwoman*, as shown in Example 17. Here, the context provides pronouns inflected for the female gender; however, gBERT still predicts the masculine gender after finetuning.

(17) Die Chairwoman hat ihren Stift fallen lassen.
the chairwoman has her pen fall let
DEF.FEM.SG chairwoman AUX.PRS.PRF POSS.3SG.F pen fall let.PTCP

"The chairwoman let her pen fall."

Other Lexical Predictions

A final error category consists of other German words predicted instead of determiners (24 loanwords, all in zero-shot). Most of these are produced by mBERT (13), followed by germanBERT (8), dbmdzBERT (4), and gBERT (4).

Frequently predicted words include *auch* (English: "also"), *als* (English: "when"), and *für* (English: "for"). Additionally, a few multi-word expressions appear, such as *Song contest*, *Public Healthiness*, and *Baby Boomer*. These errors are likely due to patterns in the pretraining data. That they only occurred in the zero-shot setting indicates that the models learned to converge to the constraints of predicting determiners after finetuning.

Most of these errors do not form grammatical sentences. However, there are some exceptions, for example, *Activity* can refer both to an activity or to the game, making the prediction in example 18 a grammatical sentence, though not inferring gender.

(18) Und Activity macht Spaß
and Activity make.3SG.PRS fun
"And Activity is fun."

Another interesting error is that mBERT predicted only a subtoken for one sentence. Example 19 shows the original masked sentence, where the expected determiner was *der* (masculine). Example 20 shows that mBERT predicted the subtoken *##er* instead of a determiner, where the subtoken *##er* is attached to *uns*, forming the possessive determiner *unser* (English: "our") and altering the syntactic structure of the sentence. The resulting form *unser* is typically associated with masculine agreement, which may indicate that the model treats *Insider* as masculine in this context. This is the only instance of this kind.

(19) Die Infos hat uns [MASK] Insider gegeben.
the.PL info.PL have.3SG.PRS us DET insider given
"An insider gave us the information."

(20) Die Infos hat uns ##er Insider gegeben.
die.PL info.PL have.3SG.PRS 1PL.DAT POSS.1PL insider give.PTCP
Intended: "An insider gave us the information."

6.3 Loanwords With Multiple Genders

The dataset included 17 instances of loanwords that take more than one gender, of which eight occurred in the test set. The gold gender was the gender associated with the subclass under which the loanword occurred. For three of the instances in the test set, at least one model predicted the alternative gender; for all others, the gold determiner was predicted.

6.3.1 Sample Cases

The first loanword for which the alternative gender was predicted is *Bit* (masculine or neuter, gold: masculine). Before finetuning, all monolingual German models predict the masculine definite nominative determiner *der* (mBERT predicts a feminine determiner). After finetuning, the monolingual models change their predictions to *das* (neuter). Both predictions are grammatically correct, while the finetuned prediction does not match the gold label.

For *Pixel* (masculine or neuter, gold: neuter *das*), gBERT and mBERT consistently predict the masculine definite nominative in both settings. GermanBERT predicts the neuter determiner *das* in the zero-shot setting but switches to the masculine determiner *der* after fine-tuning.

For File (masculine or neuter, gold: masculine *der*), most models predict the neuter gender in both zero-shot and fine-tuned settings (mBERT predicts the feminine gender). This suggests a stronger learned preference for this form, potentially reflecting patterns in the training data. This will be further examined in the following section with reference to L1 German speaker data.

Overall, most instances of loanwords with multiple gender assignments are either predicted correctly according to the gold label or align with one of the valid gender

options when alternatives are considered. This indicates that the models generally capture gender variability in these cases, though with different preferences across models. mBERT shows a tendency to default to neuter in ambiguous cases after fine-tuning and is the only model producing occasional, clearly incorrect feminine forms.

6.3.2 Comparison to Human Annotations

The survey collecting gender judgments from L1 speakers included all loanwords, including the same three loanwords analysed in the previous section: (*Bit*, *Pixel*, and *File*). This allows a direct comparison between human preferences and model predictions. For each loanword, the responses of ten participants were collected.

The participants were aged 18 or older and had lived in Germany until the age of 18, with German as their L1 language. All participants attended bilingual secondary education approximately five years ago and have therefore had high exposure to the English language. Participants now live in different regions of Germany and Austria, including northern, eastern, western and southern Germany as well as Vienna.

For *Pixel*, eight out of ten participants selected the masculine gender, which aligns with most model predictions in both settings. The human preferences are particularly consistent with the predictions of the finetuned models, as most of them converge on the masculine gender, despite the gold label being neuter.

The loanword *File* shows the most ambiguous human annotation pattern, with five feminine, four neuter, and one masculine response. In German (according to the Duden), only the masculine or the neuter gender is acceptable. This ambiguity is reflected in the model behaviour, where most models (except mBERT in the zero-shot setting) consistently predict the neuter gender, as noted in the previous section. The masculine determiner is never predicted.

For the loanword *Bit*, the majority of human annotators chose the neuter determiner *das* (7/10 annotations), three annotators chose the masculine determiner *der*, while none chose the feminine determiner. This is not the gold determiner, but not incorrect. This preference is also exhibited by mBERT after finetuning (see Section 6.3.1).

For *Lift*, human annotators predominantly select the masculine gender (10/10 predictions). This is also reflected in the model predictions after fine-tuning, where both loanwords are mostly classified as masculine. However, in the case of *Lift*, mBERT and germanBERT switch to the neuter gender after fine-tuning, despite the presence of phonological patterns in the training data that would support masculine assignment, as discussed in the previous section.

Overall, the survey suggests that L1 speakers show some variability in gender assignment across loanwords, particularly for ambiguous cases such as *Essay* and *File*. Compared to this, the models tend to align more closely with the majority human judgment after fine-tuning, except for the loanword *File*. However, differences remain for specific cases, for example *Lift*, where model behaviour does not fully align with the human preferences.

6.4 Additional Influential Factors

This section examines additional factors that may have influenced the models' predictions, including loanword length, tokenisation patterns, loanword frequency, and distance to the loanword. Performance is evaluated using accuracy scores, where a

correct prediction is one that correctly identifies a loanword's gender, either through the gold determiner or a valid alternative.

6.4.1 Loanword Length

The length of the loanword is the number of characters it is composed of. This section analyses whether loanword length influences the models' ability to correctly infer gender. The corresponding accuracy by length is shown in Table 6.11.

	mBERT		germanBERT		gBERT		dbmzBERT	
	Zero	Fine	Zero	Fine	Zero	Fine	Zero	Fine
3	60.00	100.00	40.00	100.00	40.00	80.00	40.00	80.00
4	62.96	92.59	77.78	92.59	85.19	88.89	88.89	96.30
5	50.00	76.67	70.00	76.67	90.00	90.00	73.33	90.00
6	71.43	85.71	71.43	85.71	85.71	96.43	82.14	100.00
7	58.33	97.22	75.00	97.22	75.00	97.22	72.22	94.44
8	68.97	82.76	72.41	82.76	79.31	100.00	82.76	100.00
9	56.00	92.00	72.00	92.00	84.00	92.00	68.00	92.00
10	87.50	100.00	68.75	100.00	75.00	87.50	87.50	93.75
11	64.29	92.86	64.29	92.86	64.29	92.86	71.43	92.86
12	60.00	100.00	100.00	100.00	90.00	100.00	90.00	100.00
13	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
14	100.00	100.00	100.00	100.00	66.67	66.67	66.67	66.67
16	0.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00

Table 6.11: Model Performance by Loanword Length (Accuracy Taking Alternative Predictions into Account)

Most loanwords in the dataset have a length of 5–8 letters, with an average length of 7.3 letters. Overall, shorter loanwords (3–9 letters) show lower accuracy across models (approximately 60–70%) compared to longer loanwords (10–14 letters), which reach accuracies between 70% and 100%.

This pattern differs slightly between German-specific models and the multilingual model. The monolingual German models already achieve relatively high accuracy for shorter loanwords, suggesting that they have learned German lexical and orthographic regularities sufficiently well to correctly classify even short items.

After fine-tuning, the differences in performance across loanword length decrease, particularly for mBERT. This suggests that fine-tuning helps models to learn task-specific features. It should also be noted that longer loanwords are more likely to belong to the morphological mapping category, which is highly generalisable due to its strong suffix patterns. In contrast, monosyllabic loanwords often lack such explicit morphological cues, which contributes to higher error rates for shorter items.

6.4.2 Tokenisation Patterns

The tokenisation patterns of the four models show a high overlap. mBERT and germanBERT tokenise all loanwords in the same way, and gBERT and dbmdzBERT also share the same tokenisation pattern across all loanwords. When comparing the error patterns and the overall accuracy results, models that follow the same tokenisation patterns also tend to classify loanwords in the same way, assign the same determiner,

and tend to fail on the same loanwords. This suggests that subtokenisation has a strong effect on model performance and accuracy.

The average tokenisation length by category can be seen in Table 6.12. The table shows that gBERT and dbmdzBERT tend to split loanwords into more subtokens. This suggests that a higher degree of subtokenisation may be associated with better gender prediction, as it potentially allows the model to capture more fine-grained morphological information.

	mBERT & germanBERT	gBERT & dbmdzBERT	Samples
Biological Gender	2.33	2.50	48
Phonological	1.52	1.87	46
Translation	1.74	2.03	61
Morphological	2.41	2.68	74

Table 6.12: Average Loanword Length Across Models and Classes

The subtokenisation patterns are particularly interesting for the morphological mapping category, where suffixes correspond directly to gender. Table 6.13 shows how often each suffix in the morphological mapping category was its own suffix across models.

	mBERT & germanBERT	gBERT & dbmdzBERT
-ing	7	10
-ion	1	3
-er	12	19
-ness	5	5
-ity	9	10

Table 6.13: Correct Predictions per Subtokenised Suffix After Finetuning

When evaluating performance for correct gender assignment (while accounting for alternative predictions and excluding plural forms), representing suffixes from the morphological category, as well as gender-marking suffixes from the biological male and female categories as individual subtokens, increases the likelihood of correct gender prediction. This is shown by higher performance for individually subtokenised suffixes in Table 6.14, where the gender prediction for loanwords with individually subtokenised suffixes consistently outperforms that for loanwords which do not have individually subtokenised suffixes (with the exception of germanBERT in the zero-shot setting).

	mBERT		germanBERT		gBERT		dbmdzBERT	
	Zero	Fine	Zero	Fine	Zero	Fine	Zero	Fine
Individual	68.18	100.00	68.97	100.00	87.88	100.00	87.88	100.00
Other	61.54	92.31	86.67	92.31	85.37	92.68	78.05	92.68

Table 6.14: Performance by Tokenisation Type (Individual or Other) across Models (accuracy %)

6.4.3 Loanword Frequency

To analyse loanword frequency, a frequency list based on parts of the pretraining data of two German models (gBERT and dbmdzBERT) was used. This list is derived from the German 300K NewsCrawl frequency data. As NewsCrawl is only part of the pretraining data for gBERT and dbmdzBERT, the frequency ranks presented in this section

only reflect actual pretraining exposure for these two models. For mBERT and germanBERT, the same ranks serve only as a general proxy for how common a loanword is in German, rather than as a measure of how often each model encountered it during pretraining.

Each loanword in the dataset was matched against this frequency list, and a separate frequency list was then created containing only the loanwords included in the analysis. Based on this, five frequency thresholds were defined, grouping the loanwords into categories ranging from very low to very high frequency.

A very low frequency corresponds to a one-time occurrence, a low frequency to three occurrences, an average frequency to six occurrences, a high frequency to sixteen occurrences, and a very high frequency to 616 occurrences. The distribution of loanword frequencies across the test set is shown in Table 6.15.

CLASS	Very Low	Low	Average	High	Very High	Unknown
Phonological	6 (13.04%)	10 (21.74%)	5 (10.87%)	7 (15.22%)	11 (23.91%)	7 (15.22%)
Plural	4 (20.00%)	4 (20.00%)	1 (5.00%)	2 (10.00%)	4 (20.00%)	5 (25.00%)
Translation	10 (16.39%)	11 (18.03%)	3 (4.92%)	17 (27.87%)	10 (16.39%)	10 (16.39%)
Biological	8 (16.67%)	11 (22.92%)	5 (10.42%)	2 (4.17%)	2 (4.17%)	20 (41.67%)
Morpho M.	14 (18.92%)	7 (9.46%)	5 (6.76%)	9 (12.16%)	7 (9.46%)	32 (43.24%)

Table 6.15: Number of Loanwords per Frequency Category and Class. Percentages in Parentheses Indicate the Proportion within each Class

Table 6.15 shows that loanword frequency is not equally distributed across classes in the dataset. Additionally, we can look at the performance by frequency per model (Table 6.16). It shows that a higher frequency is not indicative of better performance and that models can also perform well in assigning gender to loanwords that may not have been seen during pretraining.

	mBERT		germanBERT		gBERT		dbmdzBERT	
Frequency	Zero	Fine	Zero	Fine	Zero	Fine	Zero	Fine
Very Low	69.64	100.00	83.33	100.00	84.52	100.00	86.31	100.00
Low	61.11	84.38	80.00	87.50	87.50	93.75	86.67	100.00
Average	69.70	78.79	88.89	95.45	89.10	92.02	85.71	89.47
High	58.33	77.08	68.75	83.33	75.00	85.42	72.92	85.42
Very High	69.64	86.31	83.33	93.75	87.12	93.94	88.64	95.45
Unknown	66.15	83.85	69.23	88.46	72.31	90.00	67.69	87.69

Table 6.16: Model Performance Across Frequency Classes (accuracy %) in Zero-Shot and Finetuned Settings

6.5 Additional Tests

As mentioned in the dataset description section, there were two additional test sets to probe the model. The predictions of all four models, both before and after finetuning, will be analysed in this section. Because these tests are relatively small (ten items for the context test and between 11 and 24 items per gender for the semantic test), they only highlight tendencies rather than providing statistically robust evidence.

6.5.1 Context Test

The first additional test set probes the effect of context on several loanwords that correspond to a biological gender. This test set contains ten instances of loanwords referring to one biological gender (male or female), using them with context words typically associated with the opposite gender. For example, the loanword *Coder* (male), was used in combination with the word *Lipstick*. All of these loanwords also occurred in the original test set.

For almost all instances, all models correctly identified the grammatical gender of the loanword, even when they were used with a context that is more often associated with the opposite gender. There was one instance where mBERT originally predicted the masculine determiner *der* for the loanword *Surferin* (female) in the zero-shot setting in the context of example 21, and the context of the opposite gender changed it to have a feminine prediction (Example 22).

(21) Wie viele Wellen wird die Surferin heute reiten?
how many waves will die surfer.F today ride
how.many wave.PL AUX.3SG.FUT DET.F surfer.F today ride.INF

"How many waves will the female surfer ride today?"

(22) Wieso hat die Surferin nur eine Badehose an?
why has the surfer.F only a swimsuit.bottom on
why AUX.3SG.PRS DEF.F.SG surfer.F only INDEF.F.SG swimsuit.bottom PRT

"Why is the female surfer wearing only swimsuit bottoms?"

These results suggest that context in the form of descriptive adjectives does not challenge the models and their predictions for the correct grammatical gender remain steady.

6.5.2 Plurals

The plurals were added to the training data but are not part of the gender analysis, as they all take the same determiner regardless of grammatical gender, which prevents the determiner from being used as a proxy.

The accuracy scores by models and settings are reported in Table 6.17. All models show an initially very strong performance in the zero-shot setting. After finetuning, all models achieve ceiling performance for plural loanwords.

	mBERT		germanBERT		gBERT		dbmdzBERT	
Plural	Zero	Fine	Zero	Fine	Zero	Fine	Zero	Fine
Plural	75.00	100.00	90.00	100.00	100.00	100.00	100.00	100.00

Table 6.17: Correct Predictions for Plural Class in Zero-Shot and Finetuned Settings

The high performance on plural loanwords across all models, both before and after finetuning, suggests that models are sensitive to number agreement cues in the surrounding context, such as plurally inflected verbs. Additionally, most plural English loanwords use the English plural marker *-s*, which may be a highly exploitable cue for

the models. This marking is often reflected in the subword tokenisation, where the *-s* suffix often is its own subtoken, making plural forms more easily recognisable across contexts.

To further investigate the model's performance on plural nouns, an additional test set was created containing the singular counterparts of 16 of the 20 plural instances from the original test set. In this test set, the original plural loanwords were converted to their singular forms, allowing the predicted determiner to serve as a proxy for grammatical gender assignment. Four items had to be excluded because they lack a conventional singular form (for example *the news*).

The models performance on the test set with the singular versions of plural nouns shows that for most models, the gender prediction abilities decreased after finetuning. However, there was no overgeneralisation to the feminine determiner *die* for the singular cases of these loanwords. Apart from this, gender assignment for grammatically masculine nouns seems to be more stable.

	mBERT		germanBERT		gBERT		dbmdzBERT	
Category	Zero	Fine	Zero	Fine	Zero	Fine	Zero	Fine
Overall Accuracy	75.00	62.50	100.00	68.75	87.50	93.75	100.00	81.25
Masculine	100.00	100.00	100.00	85.71	85.71	100.00	100.00	71.43
Feminine	100.00	50.00	100.00	50.00	50.00	50.00	100.00	50.00
Neuter	42.86	28.57	28.57	57.14	100.00	100.00	100.00	100.00

Table 6.18: Accuracy (%) on Plural Test

These results suggest that correct determiner assignment for plural nouns is primarily driven by contextual cues, such as agreement with plural verb forms, rather than by coincidentally strong performance on the specific set of plural loanwords included in the original test set.

6.5.3 Semantic Test

The semantic test set included loanwords belonging to more loosely defined semantic categories based on the *Gattungsprinzip*, according to which grammatical gender is assigned on the basis of an overarching conceptual category rather than formal linguistic cues. This test set includes three semantic subclasses, each associated with one grammatical gender. The first subclass, natural elements, includes loanwords referring to wind and precipitation, which are typically assigned masculine gender, as discussed in the literature review. The second subclass consists of technology-related loanwords that are associated with the broader German concept *das Gerät* (English: "the device") and therefore tend to be neuter. The third subclass contains event-related loanwords that are linked to the semantic category *die Veranstaltung* (English: "the event") and are therefore assigned the feminine gender.

Because these semantic categories are motivated by broader conceptual associations rather than by productive and systematic cues, such as morphological suffixes, they have weaker gender-assignment regularities than the other subclasses included during training. Consequently, they were incorporated only as an additional evaluation set, which allows for assessing the model's abilities to generalise to semantically motivated categories that were not represented during the training process.

Accuracy by Class and Gender

The model's performance in both the zero-shot and the finetuned setting is presented in Table 6.19. These scores show the accuracy in inferring gender, meaning it takes possible alternatives provided in Table 5.1 into account.

	mBERT		germanBERT		gBERT		dbmdzBERT	
Gender	Zero	Fine	Zero	Fine	Zero	Fine	Zero	Fine
Natural Elements (M)	100.00	100.00	85.71	100.00	100.00	100.00	100.00	100.00
Events (F)	52.17	52.17	91.30	56.52	91.30	91.30	95.65	95.65
Technology (N)	56.25	62.50	100.00	87.50	100.00	87.50	81.25	81.25

Table 6.19: Accuracy (%) on Semantic Test by Grammatical Gender in Zero-Shot and Finetuned Settings

For the masculine gender category, all models except GermanBERTmodels' achieve perfect performance in the zero-shot setting, suggesting a higher exposure to loanwords of this category during pretraining. Following fine-tuning, all models reach ceiling performance in this category, likely due to the models overgeneralising to the masculine gender, as this was overrepresented during training. For the event-related category, none of the fine-tuned models outperforms their zero-shot version, and the performance of germanBERT decreases after fine-tuning. The neuter technology subcategory shows a mixed pattern: mBERT improves after fine-tuning, whereas germanBERT and gBERT exhibit lower performance after fine-tuning. In contrast, dbmdzBERT remains stable across both settings, with no observable change in performance.

Errors

Taking a further look at the errors made by the models in these subcategories suggests that finetuning had steered the models to overgeneralise to the masculine determiner *der*, especially for technology-related words. For event-related loanwords, models tend to assign both the neuter and the masculine gender relatively frequently. For the natural elements category, there is only one mistake made across all models and settings.

The results of this test suggest that for more abstractly motivated semantic categories, the monolingual German models have an initially strong idea of the grammatical gender for an English loanword, while the multilingual models have more difficulties. Finetuning models on literature-based rules, such as morphological and phonological patterns, or stronger semantic relations such as the biological genre of translation, does not improve performance on more abstract semantic categories.

GermanBERT and gBERT do not benefit from fine-tuning when predicting gender for abstract semantic categories, as their performance decreases for at least one gender or subclass. In contrast, dbmdzBERT shows stable performance before and after fine-tuning, suggesting that it does not compromise its initially acquired knowledge of these categories when learning additional relations during fine-tuning. mBERT is the only model for which performance on the semantic test shows a slight increase after finetuning.

Chapter 7

Discussion

The discussion chapter of this thesis will summarise the most influential factors and what the models learned for gender assignment to English loanwords in German, based on the results and the analysis. Furthermore, the current research questions and sub-questions are addressed. Then, a set of drawbacks of this thesis is listed, and suggestions for future research are provided.

7.1 Hierarchy and Gender Distribution

As discussed in the previous literature, there are rankings for both a default gender assignment to English loanwords and a ranking for the rule-based patterns.

The default hierarchy by gender is masculine > feminine > neuter (Nyenhuis and Pittner, 2012). This is reflected in the predictions by the models, where masculine determiners dominate the predictions. The predictions support the default hierarchy introduced by Nyenhuis and Pittner (2012), while not aligning with Schulte-Beckhausen (2002), who suggests that the default for unseen loanwords is the neuter gender (see Section 2.3.1). Especially when looking at the additional semantic test, it becomes obvious that the masculine gender determination is easiest for models, both before and after finetuning, while most difficulties arise for the neuter categories.

The dataset distribution aligns with this hierarchy, meaning that the dominance of masculine predictions may stem from overexposure during training. This is particularly relevant for the neuter gender, which is not only predicted least accurately but is also the most underrepresented in the dataset (see Table 3.3). Its weaker performance can therefore not be attributed to a linguistic default alone. However, dominance of masculine predictions is already present in the zero-shot setting, suggesting that this reflects a linguistic default internalised during pretraining rather than a bias introduced by the additional training data. The same holds for neuter instances. The lower percentage of correct predictions is already visible before finetuning, which indicates that at least part of the difficulty is inherited from pretraining rather than caused by its underrepresentation in the training split.

Additionally, the rank of the rule-based patterns is reflected in the results after fine-tuning. The rules-based ranking for loanwords was biological gender > morphological (suffix-based) patterns > translation > phonological rules (and lastly the *Gattungsprinzip*). The performance of the models before finetuning suggests that biological gender is easiest to capture, followed by morphological mapping (for all monolingual German models). The translation-based category is hardest in the zero-shot setting.

After finetuning, classes can be ranked by performance as follows: biological gender > phonological mapping > morphological mapping > translation. Performance on instances based on the *Gattungsprinzp* is worse in comparison to the other classes; however, this class was unseen during training, and can therefore not be ranked.

7.2 Influential Factors

The results and the analysis section have demonstrated that the morphological patterns are the most influential when it comes to gender determination. These suffix-based patterns are not only strong indicators in the morphological subcategory, but also when it comes to instances of the biological gender category, which have a gender based suffix and for plurals. Translation-based and phonological instances are hardest for the four models.

7.2.1 Finetuning

Finetuning increases the performance of all models in all classes. Some categories show greater gains from finetuning than others, and mBERT seems to benefit most from the German-specific training. The German monolingual models seem to have a better internal representation of German grammar and are therefore already able to generalise across English loanwords well, based on just the pretraining.

Finetuning also makes the models converge towards the gold (nominative or accusative) determiner. When taking the alternative predictions into account, it becomes evident that they are predicted in the zero-shot setting, and all models, for almost all loanwords, change to the correct gold determiner after fine-tuning.

7.2.2 Loanword Length, Frequency and Tokenisation

Loanword length and frequency seem to have very limited and non-linear effects on performance. This is meaningful, as it suggests that models can generalise gender assignment to loanwords they have not frequently encountered during pretraining. Tokenisation patterns, however, seem to have an influence.

For words that had a suffix marking (instances of the morphological category, some biological gender instances and plurals), the gender was more often predicted correctly if the suffix was a separate subtoken. This suggests that consistent subtokenisation of suffixes influences the model's ability to generalise on them, and that an unseen loanword can be classified correctly when the suffix is its own subtoken. A possible reason for this is that, when the suffix is its own subtoken, the model can attend to it directly and link it to a gender, whereas a suffix that is part of a longer subtoken (than just the suffix) is harder to isolate as a gender cue.

7.2.3 Context

The very high results for the plural section also indicate that the context influences the model's predictions. Here, the gender can likely be inferred from the inflexion of the verbs, rather than just the loanword itself.

The context of descriptive adjectives, as examined in the first additional test, does not appear to influence the models' predictions, suggesting that they have learned grammatical relationships related to gender and number more strongly than societal

gender associations and biases; these are not strong enough to override morphological and lexical gender cues. Through this aligns with the findings by Omrani Sabbaghi and Caliskan (2022), who argue that grammatical rather than social gender signals dominate the embedding space; the current findings are based on a small test limited in size.

This connects the previous research on gender bias in word embeddings introduced in section 2.5. Despite the models being pretrained on large text corpora that likely contain social gender biases.

7.3 Research Question and Hypothesis

The research question for this thesis is whether finetuning can improve the internalised knowledge that different bidirectional encoder models have for the gender prediction of English loanwords in German sentences. The results and analysis show that finetuning increases the number of correct predictions for all models and most categories. It also reduces the variability across subclasses. However, finetuning does not uniformly improve all linguistic signals; suffix-based categories show a high increase in performance, while other categories, such as translation-based or phonological subclasses, remain weaker. Lastly, finetuning shifts the predictions towards the gold determiners (nominative or accusative).

The hypothesis that multilingual models outperform monolingual models is rejected. The monolingual German models mostly outperform mBERT across subclasses in both settings. However, as only one multilingual model was evaluated, further research including a broader range of multilingual and monolingual models is required before concluding that language-specific pretraining generally outperforms multilingual pretraining for this task (see Section 7.4). In the current setup, the monolingual models outperform mBERT in almost all categories and settings. This could likely be explained by mBERTs pretraining, which consists of 104 languages, where high-resource languages dominate the representations, making it harder for fine-grained German morphological patterns to be encoded. This is supported by the fact that mBERT benefits most from finetuning across all models, suggesting its pretrained representations were least aligned with the task. Additionally, mBERT is the largest model in terms of parameters. However, the smaller monolingual models still outperform it. This suggests that language-specific pretraining is a stronger predictor of accuracy for grammatical gender assignment for German loanwords.

Furthermore, mBERT and germanBERT share the same tokenisation pattern, yet germanBERT outperforms mBERT. This suggests that the performance gap cannot be attributed to tokenisation differences, indicating an advantage due to language-specific pretraining.

The underlying patterns that the models were able to capture have already been extensively discussed. In short, suffix patterns (morphological patterns, plurals and suffixes for biological gender, if used) are easiest to capture, while phonological, semantic and translation-based patterns are harder to learn.

7.4 Drawbacks

There are several drawbacks to this study. The main set of drawbacks arises from the dataset. Having a manually created dataset limited the size of the training data

available for the models. However, the models still seemed to be able to generalise on multiple patterns. Furthermore, four grammatical errors were found in the dataset during the analysis (such as the missing comma, which the models then predicted). Even though the manually created dataset allowed for the control of many factors, such as limiting hints towards a gender (through pronouns or adjectives), approximately 35 instances included these hints. Lastly, not all genders are represented equally across the dataset, as masculine loanwords dominated the training set.

An additional shortcoming is that the *Duden* was used to determine the gold gender for most loanwords in the dataset. The gender assigned by the *Duden* is assumed to reflect the overall use of the German language by L1 speakers. However, the survey asking L1 speakers to choose the preferred gender for English loanwords that take more than one gender showed that L1 speakers' judgment does not always agree with the *Duden* (see *File* example in Section 6.3.2).

Another drawback is that there is only one multilingual model included in this setup. For a fair comparison, it might have been better to include an equal number of monolingual and multilingual models, as well as maybe a bilingual model. This is part of what could be done for future research, which will be discussed in the following section.

The semantic category *Gattungprinzip* was excluded from the training data due to the limited availability of loanwords with a clear gender assignment by the *Duden* and the loosely defined nature of the semantic associations, which do not provide as generalisable rules as the morphological or phonological patterns do. Including it in training would have introduced noise rather than a generalisable signal. Consequently, it was included only as an additional test set, allowing it to serve as a trial for the model's generalisation abilities to abstractly motivated categories.

7.5 Future Research

There are several ways in which future research can extend the current approach. These will be described in the current section.

The first addition would be a more diverse training set. This training set could include more examples of each category, as well as gathering examples reflecting the natural integration of English loanwords into German sentences. This could represent the generalisations L1 speakers make more effectively, and may lead to better-performing models. Given that approximately 10% of loanwords in the current dataset required rule-based gold label assignment due to missing Duden entries, a larger dataset would also reduce the reliance on this less reliable annotation method. As a second step, examples of different cases could be included to evaluate model performance under varying grammatical conditions.

Additionally, the training process could include training more multilingual models and bilingual models. Given that mBERT was the only multilingual model evaluated, it is unclear whether the underperformance of multilingual models is a general trend or specific to mBERT's architecture and training data. Furthermore, a cross-fold validation could be done to see if the performance trends hold across different data splits.

The analysis could be extended by adding more specific test sets, such as a bigger test set of minimal pairs, in relation to biological gender. Especially minimal pairs in which one loanword can refer to both a male or a female individual would allow for a more systematic investigation of the extent to which the models rely on contextual

information when assigning grammatical gender. Additionally, another test set could be introduced that tests gender assignment for loanwords that take more than one grammatical gender. This would allow for an investigation of whether models show stable or variable predictions in cases of lexical gender ambiguity.

Another test set could investigate gender assignment for loanwords without the context of a sentence, as bidirectional encoders use the context of the entire sentence to determine a grammatical gender. This could be especially interesting for the determiner prediction of plural loanwords. Though the determiners of plurals do not reflect grammatical gender, they would allow for the investigation of the role of context. Furthermore, a separate dataset could be made that controls for other factors that were investigated in the second part of the analysis. This could, for instance, involve controlling for loanword length in order to isolate the influence of frequency.

Additionally, future work could include a more detailed analysis of the top-10 predictions for each masked token. This would be particularly interesting for plural cases and classes with high levels of underspecification, where the highest-ranked prediction may not fully reflect the model's gender preferences. In such cases, lower-ranked predictions (e.g. second or third choice) may still provide useful information about which grammatical gender the model associates with the loanword, for example, when the gold determiner appears among the top-ranked alternatives.

Chapter 8

Conclusion

This thesis investigated the ability of masked language models to predict grammatical gender for English loanwords embedded in German sentences. A dataset was manually created, and each loanword was annotated with grammatical gender according to the following four rule-based categories: biological gender, morphological patterns, phonological patterns and translation. Four BERT-style models (mBERT, gBERT, germanBERT, and dbmdzBERT) were tested using a masked token prediction setup. They were evaluated on the test set in a zero-shot setting and after fine-tuning on the training set. In the results, their predictions were grouped into three classes: correct, underspecified and incorrect. For the analysis, the underspecified category was removed, and all predictions forming a grammatically correct sentence that corresponded to the gold gender were counted as correct. Their outputs were analysed across different experimental conditions, including category-based performance, frequency effects, alternative vs. gold predictions, and confidence scores

The results show that finetuning improves model performance across all models and categories, and that monolingual German models outperform the multilingual mBERT. Furthermore, the order of category difficulty mirrors the rule-based hierarchy for gender assignment to loanwords.

Overall, the results indicate that grammatical gender prediction for English loanwords in German is largely driven by morphological cues, particularly suffix-based patterns. Other patterns, such as phonological cues or translation mappings, are harder for the models to capture. This is in line with the rule-based ranking of gender assignment to loanwords.

The research questions posed in section 1.2 can be answered as follows:

RQ1 **Can fine-tuning improve the internalised knowledge that different bidirectional encoder models have for the gender prediction of English loanwords in German sentences?**

Fine-tuning improved the performance of all models, indicating that task-specific training enhances their ability to assign grammatical gender to English loanwords.

SQ1 **Do multilingual models outperform monolingual models?**

The multilingual model used in the current study does not outperform the monolingual models with regard to grammatical gender assignment for English loanwords.

SQ2 **Which underlying patterns are captured by neural networks?**

The main underlying patterns that can be captured are morphological suffix-based patterns. Categories that have some suffix marking (such as biological gender) are captured well, particularly when the suffix is represented as an individual subtoken. Phonological and translational patterns are harder to capture. This ordering is in line with the rule-based hierarchy ranking for gender assignment to loanwords.

Overall, this thesis shows that grammatical gender prediction for English loanwords in German is strongly influenced by morphological patterns, while phonological and translation-based cues are less consistently captured by the models. It was hypothesised that mBERT would benefit from its multilingual pretraining, especially for the translation-based instances; however, this was its weakest class. The results further suggest that fine-tuning improves performance based on these patterns and shifts predictions towards the definite gold determiner. While monolingual models perform best in the current setup, further research is needed to confirm whether this holds across a wider range of multilingual and monolingual models. Future work could also make use of more controlled experimental setups, for example, by extending the use of minimal pairs, to investigate the influence of individual linguistic cues in more detail.

Appendix A

Appendix

A.1 Prompts

i need you help transforming my dataset, as an example these are my first two sentences: Der Boss hat den Deal ausgehandelt. Translation - Masculine M Boss 1 1 Abteilungsleiter Der Coach hat seine Mannschaft motiviert. Translation - Masculine M Coach 2 1 Betreuer. i need it to be transformed into the following format:
#sentence_id: starting at 488, then increasing +1 from there
#category: what is says in coloumn 2 (in my example translation)
#subcategory: coloumn 3 (in my exmaple -)
#gender: coloumn 4 (in my example masculine)
#duden: coloumn 5 (my example M)
#noun: coloumn 6 (my example boss)
#example_number: coloumn 7 (my example 1)
#text: coloumn 1 (in my example: Der Boss hat den Deal ausgehandelt.
#translation: coloumn 9 (my example betreuer)
and then conll style coloumns for each token in the text, with one row per token, the ttoken count in the first coloumn, the token in the second, then a coloumn where all tokens expet the one stored under noun receive a dash (-), and then another coloumn where all tokens receive a dash. ther should be a tab between all of these. please do NOT change any of th einfor i pass you, even if there are grammar mistakes. stay true to the information i give you. show me what you think i want from ym two examples, examples should always be separated by a blank line

1 Der - -
2 Boss N -
3 hat - -
4 den - -
5 Deal - -
6 ausgehandelt - -

give your output to me in a way that i can copy (like a code box)

Weniger anzeigen

Figure A.1: Prompt to Restructure Dataset

```
#sentence_id: 488
#category: Translation
#subcategory: -
#gender: Masculine
#duden: M
#noun: Boss
#example_number: 1
#text: Der Boss hat den Deal ausgehandelt.
#translation: Abteilungsleiter

1   Der -   -
2   Boss    N   -
3   hat -   -
4   den -   -
5   Deal    -   -
6   ausgehandelt    -   -

#sentence_id: 489
#category: Translation
#subcategory: -
#gender: Masculine
#duden: M
#noun: Coach
#example_number: 2
#text: Der Coach hat seine Mannschaft motiviert.
#translation: Betreuer.

1   Der -   -
2   Coach   N   -
3   hat -   -
4   seine   -   -
5   Mannschaft  -   -
6   motiviert   -   -
```

Figure A.2: Sample Response to Restructure Dataset

A.2 Loanword Distribution

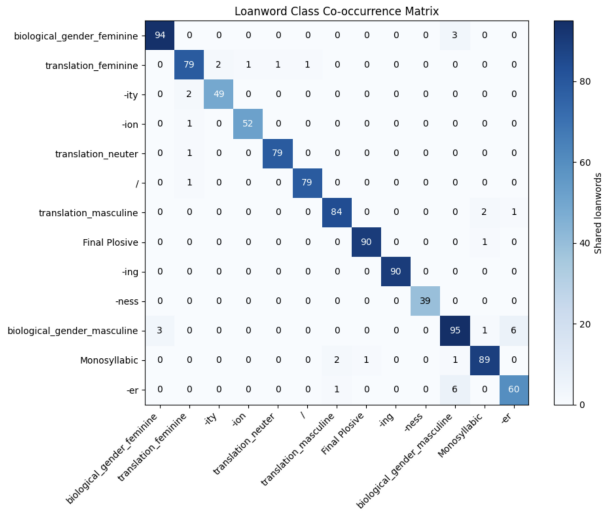


Figure A.3: Distribution of Non-Unique Loanwords

A.3 Training Configurations

A.4 Core Hyperparameters

"architecture$_f$amily": "BertForMaskedLM", "hidden$_s$ize": 768, "num$_h$idden$_l$ayers": 12, "num$_a$ttention$_h$eads": 12, "intermediate$_s$ize": 3072, "attention$_d$ropout": 0.1, "hidden$_d$ropout": 0.1, "activation$_f$unction": "gelu", "max$_p$osition$_e$mbeddings": 512, "type$_v$ocab$_s$ize": 2, "model$_t$ype": "bert", "encoder$_d$ecoder$_t$ype": "encoder"

A.5 Code Link

Link to Gitbhub: https://github.com/hannahluisee/thesis-git.git

A.6 Tables

A.6.1 SEED Results Underspecified and Incorrect

Seed	mBERT	GermanBERT	gBERT	dbmdzBERT
33	0.000	0.000	0.000	0.000
42	0.000	0.000	0.000	0.000
123	0.000	0.000	0.000	0.000
3210	0.000	0.000	0.000	0.000
Mean	0.000	0.000	0.000	0.000
Standard Deviation	±0.000	±0.000	±0.000	±0.000

Table A.1: Underspecified Prediction Rates Across Different Random Seeds

Seed	mBERT	GermanBERT	gBERT	dbmdzBERT
33	0.088	0.088	0.060	0.048
42	0.100	0.100	0.064	0.064
123	0.100	0.124	0.056	0.068
3210	0.112	0.108	0.052	0.068
Mean	0.102	0.106	0.058	0.063
Standard Deviation	±0.009	±0.012	±0.004	±0.008

Table A.2: Incorrect Prediction Rates Across Different Random Seeds

A.7 Survey

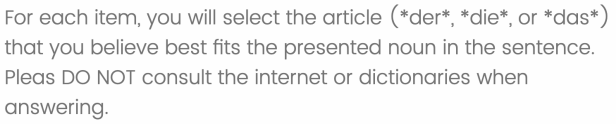


Figure A.4: Survey Instruction

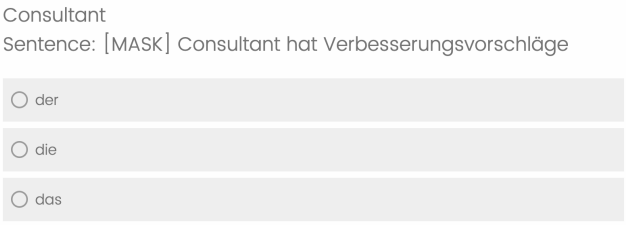


Figure A.5: Sample Instance from Survey

Appendix B

Datastatement

B.1 Purpose of the Data

This dataset was created for research purposes in relation to a master's thesis. It should aim to help answer the question whether finetuning can improve a bidirectional encoder model's ability to correctly predict the nominative determiner of an English loanword in a German sentence. The determiner is used as a proxy for grammatical gender determination.

B.2 Source

The dataset was created manually. It contains 961 unique loanwords incorporated into German sentences. Category sizes vary according to the availability of loanwords exhibiting the relevant gender assignment pattern. The majority of these loanwords are obtained from the standard German dictionary Duden. Additional loanwords were taken from the German Index of Anglicisms. The grammatical gender assigned to each loanword was verified against Duden whenever possible. The loanwords were collected and the sentences created by Hannah Altgassen in March 2026.

B.3 Dataset Description

Each loanword is incorporated into a German sentence. Loanwords were selected from Duden and the German Index of Anglicisms. For each loanword, a German sentence was manually constructed. This allowed for context to hint that a German determiner should be predicted. Context that could help indicate grammatical gender (such as inflected adjectives or relational pronouns) was kept to a minimum.

Sentences are split into five categories based on the loanword. The categories reflect patterns by which grammatical gender is assigned in German. The five categories include: morphological patterns, biological gender, phonological patterns, translation and plurals. Categories can further be divided into subcategories, each of which has its own gold determiner. Sentences were manually reviewed for grammatical correctness and consistency with the intended category and subcategory.

The dataset is available in two versions: a .csv format and a CoNLL-style file. Each loanword is incorporated into a German sentence. Each sentence has a sentence ID, a category, a subcategory, a note of all acceptable grammatical genders, a gold

determiner, and the laonword. The CoNLL-style format is also available in a 2:1:1 training, development and test split, which was created using a stratification key and random shuffling.

B.4 Scope and Representation

The grammatical gender and the loanwords themselves are largely based on the Duden. This is the standard German dictionary that reflects the general use of the German language. This means that dialectal variations are not taken into account.

B.5 Privacy and Ethics

There are no privacy concerns in the creation of this dataset. No personal data was included. Some ethical concerns are that the language usage treated as gold is based on the Duden who bases its annotations on frequency. This means that smaller language variations may not be taken into account. It does not cover loanwords from other languages.

B.6 Access and Usage

The dataset can be downloaded via GitHub: `https://github.com/hannahluisee/Thesis.git`. There are no restrictions to the usage of this data.

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